Initialisation Sets the Conditioning of LoRA Merging

SANKALP PATHAK*, Jaypee University of Engineering and Technology, India
SANJAY GARG, Jaypee University of Engineering and Technology, India
PIYUSH KUMAR SINGH, Jaypee University of Engineering and Technology, India

Background: Merging several LoRA adapters into one model sometimes costs almost nothing and sometimes destroys a task, with no cheap way to tell which in advance. LoRA starts each adapter from a random matrix, and whether the T adapters of a cohort started from one such matrix or from T independent ones is rarely reported: a single global training seed silently decides it.

Objectives: The authors set out to turn that unreported detail into a pre-merge diagnostic: a check that is run on the adapter files before any merge is attempted, that says whether a cohort has collapsed into a shared subspace, and that predicts what this implies for the merging method about to be applied to it.

Methods: The authors measure the subspace geometry of adapter cohorts across four base models under both conditions, audit 45 publicly released cohorts under a sampling frame fixed before any download, and test the consequences for five merging methods in loss and in downstream accuracy. Every test was pre-registered before the compute it governs, with the registrations and the commit ordering included so that a reader can check the sequence rather than take it on trust.

Results: Sharing the initialisation leaves the tasks in nearly the same subspace, moving the median principal cosine across four base models from 0.05 to 0.99, and 17 of the 45 public cohorts are in that state. The consequences split by method. Merges that average or sparsify task vectors are unaffected: fifteen of twenty base-by-method cells show no effect on the authors' loss metric, and none of forty show one on GSM8K or HumanEval. Merges that solve a linear system, the authors' and RegMean, run 2 to 65 times worse on a collapsed cohort, and on Mistral the merged model scores zero on GSM8K and on HumanEval; one Tikhonov term recovers both. Regularising such a solve is standard practice, but RegMean's shipped default is about 20 times too small on a collapsed cohort and applies the same dose either way.

Conclusions: Which regime a cohort is in is decided before merging begins, is invisible to the merging method, and is readable from the weight files in about a minute of CPU time without running the model. That makes it a practical pre-merge check rather than a post-hoc explanation: it is available at the moment the decision is actually taken, it is independent of which merging method follows, and on a collapsed cohort it is the difference between a working merge and a model that scores zero. The shipped default does not know which regime it is in, and the diagnostic is what tells you. Several of the authors' own earlier claims did not survive the pre-registered tests, and those outcomes are reported alongside the ones that did.

JAIR Associate Editor:

JAIR Reference Format:
Sankalp Pathak, Sanjay Garg, and Piyush Kumar Singh. 2026. Initialisation Sets the Conditioning of LoRA Merging. *Journal of Artificial Intelligence Research* 0, Article 0 (August 2026), 49 pages. DOI: 10.1613/jair.1.xxxxx

*Corresponding author.

Authors' Contact Information: Sankalp Pathak, ORCID: 0009-0006-5666-8271, pathaksankalp04@gmail.com, Department of Computer Science and Engineering, Jaypee University of Engineering and Technology, Guna, Madhya Pradesh, India; Sanjay Garg, ORCID: 0000-0002-2279-9373, gargsv@gmail.com, Department of Computer Science and Engineering, Jaypee University of Engineering and Technology, Guna, Madhya Pradesh, India; Piyush Kumar Singh, ORCID: 0009-0000-8033-3777, Piyushsingh5629@gmail.com, Department of Computer Science and Engineering, Jaypee University of Engineering and Technology, Guna, Madhya Pradesh, India.

DOI: 10.1613/jair.1.xxxxx

1 Introduction

Low-rank adaptation (Hu et al. 2022) has made it cheap to produce many specialised versions of one base model, and correspondingly attractive to fold several of them back into a single set of weights. A large family of merging heuristics now exists for this: task arithmetic (Ilharco et al. 2023), sign-election methods such as TIES (Yadav et al. 2023), stochastic sparsification such as DARE (Yu et al. 2024), subspace alignment such as KnOTS (Stoica et al. 2025), and closed-form least-squares approaches in the RegMean line (Jin et al. 2023; Nguyen et al. 2026; Salami et al. 2024).

These methods are evaluated almost entirely by benchmark averages, and the resulting picture is unstable: the same method can be best on one cohort of adapters and worst on another, with no way to tell in advance which situation one is in. This paper is about one property of a cohort that explains a good deal of that instability, and what the authors contribute is a *pre-merge diagnostic* for it: the property is measurable from the adapter weight files alone, before any merge is attempted and in about a minute of CPU time, and it predicts which family of merging methods will survive the cohort and how much regularisation a solve-based one needs on it. It is independent of which merging method is chosen afterwards, which is what makes it worth running first.

The property. LoRA factorises each update as BA, initialises A at random and B at zero, and trains. Whether the T adapters of a cohort drew *one* A or T independent ones is a detail almost never reported, and it is what a single global training seed silently decides. When the draw is shared, the tasks end up occupying nearly the same subspace. Across four base models, sharing the draw moves the median principal cosine between task row spaces from 0.047 to 0.995 (§4, Table 1). The task subspaces stop being distinguishable directions and become one direction measured four times. Nothing about this is a bug in LoRA, and nothing about it is visible in a training curve or a benchmark average.

Measuring it is cheap. The principal cosines are computed from the adapter weight files alone: no forward passes, no evaluation data, no merging, and about one minute of CPU time for a cohort of four rank-16 adapters (§4.5). This is the part of the paper the authors expect someone to use directly, and the rest of the paper is an argument that it is worth running.

The effect splits cleanly by method. The authors pre-registered the test and report both halves (§5). Across twenty base-by-method cells spanning five methods, task arithmetic, TIES, DARE, TVQ and a repaired KnOTS, fifteen show no detectable effect at the authors' 0.01 nat resolution, five favour independent initialisation and none favours shared. Scored instead on GSM8K and HumanEval, a pre-registered replication of the same comparison finds no effect in any of forty cells. The fifteen have a mechanism: each of these methods returns a bounded-norm near-mean solution that never enters the ill-conditioned directions, and its energy there is a rounding error. Of the five cells that do clear the gate, three are the subspace-alignment method, which is the heuristic most exposed to a change in subspace geometry.

Methods that solve the linear system behave differently. They reach 35 to 57 times the mean task delta with three quarters of their energy exactly in the ill-conditioned directions, and run unregularised they are 2.1 to 65 times worse on a shared-initialisation cohort, 25 to 117 times with rank truncation removed. A single Tikhonov term repairs it. The authors show this for their own encoder and for RegMean, so it is a property of the family rather than of their construction.

The obvious objection is that regularising a least-squares solve is standard practice, so the authors checked what the standard practice actually does. RegMean ships a regulariser of its own, an α-shrinkage of the off-diagonal Gram entries with a default of $\alpha = 0.9$. On a collapsed cohort it is equivalent to a ridge of about 5×10^{-4}, roughly 20 times smaller than the $\lambda^\star = 0.01$ the sweep selects, and it leaves the system at condition number 6.6 to 9.1×10^3 against 1.5×10^2 with the ridge. It applies the same dose on a well-conditioned cohort, where it is ample. That is the practical content of this paper in one sentence: **the default regularisation does not know which regime it is in, and the audit is what tells you** (§5.5).

The regime is common in released adapters. Every cohort discussed so far is one the authors trained themselves, in both conditions, deliberately. So the authors ran the same measurement over public LoRA cohorts under a sampling frame registered before any adapter was downloaded: walk the HuggingFace adapter index by downloads descending, group by publisher and base model, take the first 50 cohorts meeting criteria that are all checkable from metadata, skip none after inspection. **Seventeen of the** 45 **scorable cohorts,** 37.8%**, are in the collapsed regime** (§4.6). The distribution is bimodal rather than a continuum: one cohort of 45 lies between the two populations.

It reaches accuracy, not just the authors' metric. A large difference measured in a loss nobody deploys would still be open to the objection that the loss is the wrong unit. On the same merges, scored on GSM8K and HumanEval with the repaired scorers, the ridge recovers the shared arm past the noise gate on four of four bases on both benchmarks. On Mistral the unregularised shared-cohort merge scores **zero** on both, with all 664 generations non-empty and none discarded, and recovers to 0.46 and 0.24 (§5.8).

Why conditioning is the right quantity. The task subspaces being near-collinear does not by itself make merging impossible; T rank-r adapters with $Tr \leq d$ stay linearly independent even when nearly collinear, so an exact interpolator always exists. What changes is its norm. Under a shared draw the exact interpolator is 43 to 64 times larger than the mean task delta; under independent draws it is T, the orthogonal-subspace value, which the measurement returns to within 1.4% at both $T = 3$ and $T = 4$. The condition number of the operator $\bar{H}$ built from the task subspaces moves correspondingly, from about 1.5 to between 1.8×10^4 and 8.3×10^4 (§4, Figure 1). A method that solves for that interpolator inherits the conditioning; a method that averages does not. That is the whole of the split reported above.

The authors report the principal cosines rather than $\kappa(\bar{H})$ throughout, because the cosines are properties of the row spaces alone and hold under any curvature model, whereas $\kappa(\bar{H})$ depends on the specialisation $H_t = P_{V_t}$ adopted here (§4.4).

Where this came from. The authors arrived at the conditioning question through a rate-distortion analysis of merging, which is in the paper because it is the reason for looking at $\bar{H}$ at all and for no stronger reason than that. §3 says what it did and did not do, and Appendix A carries the formal development. In short: its floor term is exactly zero on every cohort the authors measured, its rate term rests on an assumption stated but not proved and on an imported constant, and it derives none of the four results above. Everything in §4 and §5 stands or falls on its own measurements.

What did not survive. Several pre-registered tests went against the authors, and the results below are what is left after them. The authors' own encoder loses its advantage in the move from shared-initialisation cohorts to properly initialised ones. A control the authors designed themselves withdrew their earlier headline claim that published merging benchmarks are confounded by initialisation geometry, and it stays withdrawn: a later margin-aware control would reinstate it, but that control was written after its data had been seen and the authors do not use it that way (§5.7). The rate exponent the bound predicts is unresolved: the authors' sweep lacks the dynamic range to measure it (§5.9). One of the authors' four adapters never learned its task, so the cohorts are $T = 4$ nominal and 3 effective, and the authors report every geometric quantity at both. Appendix F summarises all twelve pre-registrations and gives the commit ordering that makes them checkable; the documents themselves are in the public repository.

An earlier version of this work is public. It is posted on Zenodo under the authors' own names, titled *A Rate-Distortion Function for Model Merging* (Pathak and Garg 2026), and it predates the pre-registered replication reported here. The authors cite it rather than leave it to be found, because it still presents as headline results two things this paper does not claim: that the rate-distortion bound is the contribution, and that their encoder attains the lowest worst-task loss on all four bases at default hyperparameters. Both are superseded by the tests in §5. Where the two documents disagree, this one is current.

Contributions.

(1) **A one-CPU-minute audit** that reports, from the adapter weight files alone, whether a cohort's task subspaces have collapsed, and hence whether an unregularised solve-based merge will fail on it (§4.5).
(2) **A pre-registered prevalence estimate** for that regime in adapters the authors did not train: under a sampling frame fixed before any download, 37.8% of 45 public LoRA cohorts are collapsed, and the distribution is bimodal rather than continuous (§4.6).
(3) **A pre-registered demonstration that the collapse is invisible to standard merging heuristics and decisive for solve-based ones**, for the authors' encoder and for RegMean, at $n = 3$ cohorts in *both* regimes, with the mechanism measured: the heuristics never place more than a rounding error of their norm in the ill-conditioned directions and the unregularised solvers place three quarters of it there (§5.2, §5.3). The invisible half is measured twice, in worst-task NLL excess and in downstream accuracy, because the metric the first uses is one the authors themselves criticise: zero of forty accuracy cells show an effect, at a resolution of about six points on GSM8K and eleven on HumanEval.
(4) **A pre-registered demonstration that this reaches downstream accuracy**, so it is not an artifact of the loss the authors measure it in: on Mistral the unregularised shared-cohort merge scores zero on GSM8K and on HumanEval, with no scorer discards, and one ridge term recovers it (§5.8).

The rate-distortion bound (Theorems 4 and 5) is provenance rather than a contribution, for the reasons above. The authors also report a negative result for DARE composed with task arithmetic and the pre-registered test showing it does not extend to DARE with TIES (§5.10), which bounds an earlier claim of theirs rather than adding a new one.

2 Related Work

Model merging: constructions without limits. A wave of post-hoc merging methods followed Task Arithmetic (Ilharco et al. 2023): Fisher-weighted averaging (Matena and Raffel 2022), sparsification approaches (TIES (Yadav et al. 2023), DARE (Yu et al. 2024), DELLA (Deep et al. 2024)), rotation- and subspace-alignment families (KnOTS (Stoica et al. 2025), TSPA (Li et al. 2025), DO-Merging (Zheng et al. 2025), Core Space (Panariello et al. 2025), TARA (Jeong et al. 2026), ARM (Y. Yao et al. 2026)), and merge-time quantisation (TVQ (Kim et al. 2025), 1-bit-Merging (S. Liu et al. 2025)). All are *constructions*: each proposes an encoder and evaluates it empirically. What none of them supplies, and what this paper does supply, is a way to tell in advance which of them will work on a given cohort. The authors do also prove a rate-distortion lower bound at a storage budget, but report it as provenance rather than as a contribution, and §4.3 and §5.9 say exactly how little of it survives contact with real adapters.

Existing theoretical analyses of merging. Weight disentanglement (Ortiz-Jimenez et al. 2023) identifies the rank-r Hessian structure of LoRA fine-tunes and provides the $H_t = U_t D_t U_t^\top$ formalism the authors adopt, but gives no rate-distortion bounds. Task-Vector Bases (Zeng et al. 2025) posits a thin-Gaussian-shell geometry; ATM (Zhou, Solombrino, Crisostomi, Bucarelli, Silvestri, et al. 2024) and its successor (Zhou, Solombrino, Crisostomi, Bucarelli, D'Inverno, et al. 2025) develop a second-order Taylor bound on task-arithmetic error. None of these frames merging as lossy source coding.

Concurrent work using the same lens. Cao et al. (2026), posted while this work was in progress and independently of it, study "merging collapse" in fine-tuned LLMs, identify representational incompatibility between tasks as a strong correlate of it, and use rate-distortion theory to argue for a dimension-dependent limit on mergeability that holds regardless of the merging method. The authors therefore make no priority claim for the lens itself.

The two bounds constrain different objects, and the authors correct one of their own descriptions of the difference. An earlier draft distinguished the bound of this paper as bounding worst-task rather than average distortion. That is not a difference: their Theorem 1 bounds $\delta_{\max}(\hat{\theta}) = \max_i \mathbb{E}_X \|h(X;\hat{\theta}) - h(X;\theta_i)\|_2^2$, which is also a maximum over tasks. The real differences are these. Theirs measures distortion in the *hidden states* of a fixed layer and the authors' in *parameter* space under a per-task curvature H_t, so their quantity needs forward passes and data while the authors' is read off the adapter files. Theirs assumes linear mode connectivity, that every convex combination of the fine-tuned minima attains the same loss, and derives $\delta_{\max} \geq \frac{1}{4}\Delta^2$ with a convex merge achieving $\frac{d}{2(d+1)}\Delta^2$ by Jung's theorem, where Δ is the diameter of the task representation clusters. The authors' assumes rank-r structure and, for the rate term only, Assumption 3. And the role of the rate differs sharply: their rate-distortion statement is that $R(D) = 0$ exactly when $D \geq \Delta^2/4$ and $R(D) \geq \log_2 N$ below it, a zero-rate threshold rather than a tradeoff, whereas the content of the authors' is the R-dependence itself.

Their central negative result deserves a direct answer, because it looks like a counterexample to the authors'. They report that parameter-space conflict metrics correlate minimally with merging collapse while representational incompatibility correlates strongly, and this paper's headline instrument is a parameter-space measurement. The authors think the two agree where they overlap, for a reason the authors' own null supplies. Their bound is proved for convex merges under LMC, and convex merges are precisely the class in which the authors *also* find parameter-space geometry to have no detectable effect: fifteen of twenty base-by-method cells in NLL (§5.2) and zero of forty in downstream accuracy. The authors' positive claim is confined to merges that solve a linear system, which are not convex combinations of the fine-tuned models and which their assumption excludes. Read that way, both papers find that parameter-space quantities do not predict what averaging-type merges do; the authors add that one parameter-space quantity does predict what solve-based merges do. The authors have not tested their Δ on the cohorts measured here, and doing so would need the forward passes this paper's audit avoids, so the reconciliation is offered as an argument and not as a measurement. A reader interested in the limits of merging should read both.

Closed-form merging lineage. The authors' encoder sits in a line of closed-form merges that invert a per-task curvature surrogate. RegMean (Jin et al. 2023) solves a least-squares system in the per-task input-activation Grams (data-dependent, forward passes required); LoRM (Salami et al. 2024) carries the closed form over to LoRA modules in federated continual learning, and RegMean++ (Nguyen et al. 2026) adds cross-layer dependencies to RegMean's objective so that merged earlier layers are accounted for when merging deeper ones. The authors' differs in two ways: it is *data-free*, since the projector comes from the adapter factors alone, and it is derived from a worst-task rate-distortion lower bound rather than from an empirical fit. What it shares with that line is more important here than what separates it. All of these methods solve a linear system in a curvature surrogate, and all carry a ridge term whose role §5.2 is about. The authors test RegMean directly on both cohorts (§5.4) and find the same conditioning effect there, which is what licenses talking about the family rather than about the encoder introduced here. LoRM and RegMean++ are not run: RegMean++'s distinguishing mechanism is a cross-layer dependency in the activation Gram and LoRM's setting is federated continual learning, so neither has a data-free adapter-only form the authors could implement without inventing the part that makes it that method. The family claim therefore rests on two members of it, not five.

Rate-distortion and rotation-for-quantisation ancestry. Classical rate-distortion theory (Cover and Thomas 2006; Shannon 1959) covers the single-source setting; multi-description coding (El Gamal and Cover 1982) treats *average*, not maximum, distortion, which Lemma 1 shows is strictly weaker. The direct ancestor of the authors' achievability construction is TurboQuant (Zandieh et al. 2025), whose Theorem 3 Theorem 5 recovers at $T = 1$. Structured rotations as quantisation pre-conditioners (QuIP (Tseng et al. 2024), QuaRot (Ashkboos et al. 2024), SpinQuant (Z. Liu et al. 2025), RoLoRA (Huang et al. 2024)) motivate the authors' randomised orthogonal mixers but are single-source.

A recent negative result, and why a lower bound still matters. A large-scale empirical study (Hitit et al. 2025) reports that across four LLM families only Task Arithmetic reliably produces constructive interference, with subspace-alignment methods often failing. Their result is an *upper bound* on specific algorithms; the authors' Theorem 4 is a *lower bound* on every algorithm. The authors' measurements bear on theirs, but less directly than an earlier reading suggested. The authors find $d_{\text{eff}} = Tr$ in every layer of all four bases and in both initialisation conditions, so the task subspaces are indeed linearly independent and the irreducible floor is zero throughout (§4.3). The authors previously took that to mean subspace-alignment methods have no overlap to exploit, and cited their own KnOTS measurement as support. The authors withdraw both. Formal independence is not the absence of overlap: under shared initialisation the subspaces are near-collinear and $\bar{H}$ is ill-conditioned (§4), which is exactly the situation an alignment method might be expected to address. And the authors' KnOTS implementation reduced algebraically to Task Arithmetic under its default inner combination, so it never tested the question (§6). The authors make no claim here about published KnOTS. Where the authors' ordering differs from theirs, the difference is one of metric and setup (average interference across 16 heterogeneous benchmarks vs. worst-task NLL excess on a controlled matrix), not a contradiction.

3 Problem Setup

Task vectors and per-task geometry. Fix integers $T \geq 1$, $d \geq 2$, $1 \leq r \leq d$, and a radius $B > 0$. Each of the T tasks is specified by a pair $(\tau_t, H_t) \in \mathbb{R}^d \times \mathbb{R}^{d\times d}$ with

$$H_t \succeq 0, \quad \operatorname{rank}(H_t) \leq r, \quad \tau_t \in V_t := \operatorname{range}(H_t), \quad \tau_t^\top H_t \tau_t \leq B^2. \tag{1}$$

Per-task loss is the quadratic $f_t(w) := \frac{1}{2}(w - \tau_t)^\top H_t (w - \tau_t)$. The authors write $\boldsymbol{\tau} := (\tau_t)_{t=1}^T$, $\boldsymbol{H} := (H_t)_{t=1}^T$, and call tuples satisfying (1) *admissible*. The specialisation $H_t = P_{V_t}$ recovers the weight-disentangled LoRA setting of Ortiz-Jimenez et al. (2023); general $H_t \succeq 0$ subsumes Fisher-metric surrogates for cross-entropy.

Worst-task distortion. A merged model $w^\star \in \mathbb{R}^d$ incurs per-task distortion $D_t(w^\star) := (w^\star - \tau_t)^\top H_t (w^\star - \tau_t)$, and the figure of merit throughout is the *worst-task* distortion

$$\Delta(w^\star; \boldsymbol{\tau}, \boldsymbol{H}) \ := \ \max_{t\in[T]} D_t(w^\star), \tag{2}$$

because a merged model that collapses on a single task is not deployable. Empirically the authors report the same quantity as worst-task NLL excess: the largest, over tasks, of the merged model's negative log-likelihood on a task minus that of the task's own adapter.

Key derived quantities. Two quantities carry the rest of the paper:

$$\bar{H} := \frac{1}{T}\sum_{t=1}^{T} H_t, \qquad \bar{\tau}_H := \bar{H}^+ \cdot \frac{1}{T}\sum_{t=1}^{T} H_t \tau_t, \qquad d_{\text{eff}} := \operatorname{rank}\Big(\sum_{t=1}^{T} P_{V_t}\Big), \tag{3}$$

with $(\cdot)^+$ the Moore–Penrose pseudoinverse. The *H_t-weighted centroid* $\bar{\tau}_H$ is the unique element of $\operatorname{range}(\bar{H}) = V_1 + \cdots + V_T$ with $\bar{H}\bar{\tau}_H = T^{-1}\sum_t H_t \tau_t$, reducing to the arithmetic mean when $H_t = I_d$. The *effective dimension* $d_{\text{eff}} \leq \min(Tr, d)$ counts the independent directions a merge must reconstruct.

$\bar{H}$ is the object this paper is about. §4 shows that whether the T adapters of a cohort shared one initialisation determines the conditioning of $\bar{H}$, across four orders of magnitude, and §5 shows which merging methods that conditioning reaches.

A note on $H_t = P_{V_t}$. Every $\kappa(\bar{H})$ the authors report adopts this specialisation, so it is an index of subspace overlap under a stated curvature model and would take a different value under a Fisher-weighted H_t. This is why the authors lead on principal cosines, which are properties of the row spaces alone and hold under any curvature model (§4.4).

Where $\bar{H}$ came from. The authors arrived at this operator through a rate-distortion analysis of merging, which treats the merge as lossy compression of T task vectors into one weight vector and asks what worst-case distortion is achievable at a given storage budget. Appendix A states the resulting lower bound and a matching construction, and Appendix B proves them. The authors report the analysis as provenance rather than as a contribution, for three reasons given in full there and summarised here: its floor term is exactly zero on all four base models under every initialisation tried here, so it carries no operational content for LoRA merging as practised (§4.3); its rate term rests on an assumption stated but not proved and on a constant imported from TurboQuant (Zandieh et al. 2025), and the attempt made here to measure the exponent it predicts has too little dynamic range to do so (Appendix E); and it constrains a merged model carrying no rank constraint, while every merge the authors run is truncated to rank r (§4.4). What it did do is put the geometry of the union of the task subspaces at the centre of the problem, which is what sent the authors to measure principal cosines. Nothing else the authors tried pointed there, and everything in §4 and §5 stands or falls on its own measurements.

4 Two Regimes: What Initialisation Controls

Theorem 4 bounds worst-task excess below by $B^2(1 - d_{\text{eff}}/(Tr))$, which vanishes exactly when the T task subspaces are in direct sum. This section reports what that quantity does on real LoRA cohorts. The answer is that it vanishes, which is why this section is organised around conditioning rather than around the floor.

4.1 Why shared initialisation collapses the task subspaces

In the reference LoRA implementation (Hu et al. 2022) an adapter is $\Delta W = BA$ with A randomly initialised and B initialised to zero. Two consequences matter. Because $B = 0$ at the start, the gradient with respect to A is proportional to B and vanishes at initialisation, so A moves comparatively little over training. That second step is a measurement rather than an inference: on a cohort trained with the checkpoints to answer it, A ends training about 13% of its own norm away from where it started (Appendix D.4). And in most pipelines the initialisation is drawn from a single global seed, so if the T adapters of a cohort are trained with that seed, as is natural when the seed is meant to control run-to-run reproducibility, all T begin from an *identical* A.

Since $\text{range}(\Delta W_t^\top) \subseteq \text{range}(A_t^\top)$, if every A_t is a small perturbation of one shared matrix then the task subspaces are near-collinear. This is not a statement about the tasks being related; it is an artifact of initialisation and survives when the tasks are unrelated.

The same experiment separates the two candidate causes, which is what makes it worth running rather than merely reassuring. A shared-initialisation cohort and an independently initialised one drift by 0.1308 and 0.1303, a difference of 0.0005, and end with median principal cosines of 0.996 and 0.047. Training displaces A by the same amount in both conditions, so the collapse is attributable to where the adapters started and not to anything training does to them.

4.2 Measurement

The authors train $T = 4$ adapters at rank $r = 16$ ($Tr = 64$) on four base models (Llama-3.1-8B-Instruct, Mistral-7B-v0.3, Qwen2.5-7B-Instruct, Yi-1.5-9B-Chat) for four unrelated tasks: grade-school mathematics (`gsm8k`), instruction following (`alpaca`), code generation (`magicoder`), and translation (`flores`). Two conditions: a *shared* cohort using one global seed, and *independent* cohorts drawing a separate initialisation per task, repeated three times

Table 1. Subspace geometry, $T = 4$, $r = 16$, $Tr = 64$, `indep1` shown for the independent arm; `indep2` and `indep3` agree to three decimals (Appendix D). Hard d_{eff} is at $\epsilon = 10^{-1}$.

base	init	med. cosine	$\sigma_{\max}$	soft d_{eff}	hard d_{eff}	floor at 10^{-6} / 10^{-2}
Llama-3.1-8B	shared	0.9948	1.999	16.33	20.8	0 / 0.324
	independent	0.0473	1.107	63.23	64.0	0 / 0
Mistral-7B	shared	0.9947	1.999	16.28	19.9	0 / 0.365
	independent	0.0468	1.106	63.24	64.0	0 / 0
Qwen2.5-7B	shared	0.9964	1.999	16.35	21.4	0 / 0.161
	independent	0.0500	1.131	63.08	64.0	0 / 0
Yi-1.5-9B	shared	0.9973	2.000	16.28	20.2	0 / 0.230
	independent	0.0467	1.102	63.25	64.0	0 / 0

(`indep1/2/3`). The translation adapter does not learn its task; the authors report that defect and its consequences in §4.7 and carry the qualification throughout.

For each cohort the authors compute, per layer and averaged over eight sampled layers, the median principal cosine between the T task row spaces; the top singular value $\sigma_{\max}$ of the stacked bases, which is $\sqrt{T}$ under exact collinearity and near 1 under orthogonality; and two notions of effective dimension.

Two effective dimensions, and which one the theory uses. With $M = [V_1 \cdots V_T]$ the concatenated orthonormal bases and σ_i its singular values, the *hard* effective dimension at tolerance ϵ is $|\{i : \sigma_i > \epsilon\, \sigma_1\}|$, and the *soft* effective dimension is the participation ratio

$$d_{\text{eff}}^{\text{soft}} := \frac{\left(\sum_i \sigma_i^2\right)^2}{\sum_i \sigma_i^4}. \tag{4}$$

The quantity in Lemma 2 is a *rank*, so it is the hard one at any tolerance small enough to be numerically meaningful. The soft version is a continuous diagnostic that degrades gracefully instead of jumping, which makes it convenient to report, but it is **not** the quantity in the theorem, and substituting it into $B^2(1 - d_{\text{eff}}/(Tr))$ is not licensed. §4.3 shows what that substitution costs: it is the single largest numerical error the authors have had to correct.

The geometric separation is large and reproducible. Under shared initialisation the median principal cosine is above 0.994 and $\sigma_{\max}$ is 1.999 against a collinear maximum of $\sqrt{T} = 2$; under independent initialisation the cosines are near 0.047 and $\sigma_{\max}$ near 1.1. The three independent cohorts agree to three decimal places on every column, so this is structural rather than a sampling accident.

4.3 The irreducible floor is zero in both regimes

It is tempting to read the soft column of Table 1 through Lemma 2 and conclude that the shared cohort carries an irreducible floor of $B^2(1 - 16.3/64) = 0.745B^2$ against $0.012B^2$ for the independent one. That is wrong, and the rest of this subsection is why.

The correct statement has two halves, and separating them matters because one is algebra and the other is not.

Algebraically, the floor is zero for every cohort here. By Lemma 2 it vanishes iff the V_t are in direct sum, and T rank-r adapters with $Tr \leq d$ span independently unless they coincide exactly. Near-collinear is not collinear:

Table 2. Conditioning of $\bar{H}$ and the norm of the exact interpolator, relative to the mean task delta. The independent-arm amplification returns T, the value for exactly orthogonal subspaces, to within 1.1% in the worst of the four cells shown, all of which are $T = 4$; across both values of T the worst is 1.4% (Appendix D.2).

base	$\kappa(\bar{H})$		$\|\bar{\tau}_H\|/\|\bar{\Delta}\|$	
	shared	independent	shared	independent
Llama-3.1-8B	1.76×10^4	1.5	48.3	4.0
Mistral-7B	3.21×10^4	1.5	52.3	4.0
Qwen2.5-7B	2.42×10^4	1.6	43.4	4.0
Yi-1.5-9B	8.30×10^4	1.5	64.1	4.0

even at median cosine 0.995 the four 16-dimensional row spaces are linearly independent, $d_{\text{eff}} = Tr = 64$, and an exact interpolator exists. This is a rank statement and needs no measurement.

Numerically, that is not a property one can verify by evaluating (11), and the authors do not claim to have done so. Computing $B^2 - \bar{\tau}_H^\top \bar{H} \bar{\tau}_H$ requires a pseudoinverse of $\bar{H}$, whose condition number reaches 8.3×10^4 on the shared arm (§4.4), so the answer depends on the rank tolerance. At rtol $= 10^{-6}$ the authors obtain zero on all eight cohort-base pairs; at 10^{-2} the shared arm returns 0.16 to 0.37 B^2 and the independent arm still returns zero. Reporting "exactly zero" for the shared arm would be reporting a tolerance choice as a fact.

That tolerance sensitivity is the finding, not an inconvenience. **A shared-initialisation cohort is numerically indistinguishable from a genuinely positive-floor one**, and only exact arithmetic separates them. So the algebraic answer, that the floor term of Theorem 4 is vacuous for LoRA cohorts in general position, is correct and also the wrong thing to look at: it is a statement about a rank that no finite-precision merge can resolve. What a practitioner experiences is governed by how far the cohort sits from the degenerate case, which is a conditioning question. §4.4 asks it directly.

One point stands regardless of tolerance: a soft surrogate must not be substituted into a formula derived for a rank. Doing so yields a 60-fold floor difference that is not the difference between two floors at any tolerance the authors have tried.

4.4 What initialisation actually controls is conditioning

A zero floor says an exact interpolator exists. It says nothing about its norm. That is where the two regimes separate.

Table 2 gives the result. The condition number of $\bar{H}$ differs by four to five orders of magnitude between the two conditions. The exact interpolator is 43 to 64 times larger than the mean task delta under shared initialisation, against 3.96 to 4.00 under independent initialisation. That value is T, the identity for exactly orthogonal subspaces, and recomputing with the degenerate adapter dropped returns 2.96 to 3.00 at $T = 3$ (Appendix D.2). Neither is fitted. The authors record this as a correctness check on the implementation and nothing more: reproducing an identity confirms the code computes what they intend, and says nothing about whether the quantity matters. §5.2 is where that is tested.

So initialisation does not set the floor. It sets how ill-conditioned the merge problem is, and therefore how large a solution must be to interpolate the tasks.

Is $\kappa(\bar{H})$ a property of the cohort or of the authors' surrogate? $\bar{H}$ is built from the LoRA factors under the specialisation $H_t = P_{V_t}$, so the question is fair and they answer it only partly. What is surrogate-independent is the geometry: the principal cosines and $\sigma_{\max}$ in Table 1 are properties of the row spaces range$(A_t^\top)$ alone, and any H_t with range$(H_t) = V_t$ inherits the near-degeneracy of $\sum_t P_{V_t}$, because that degeneracy is the overlap of the V_t

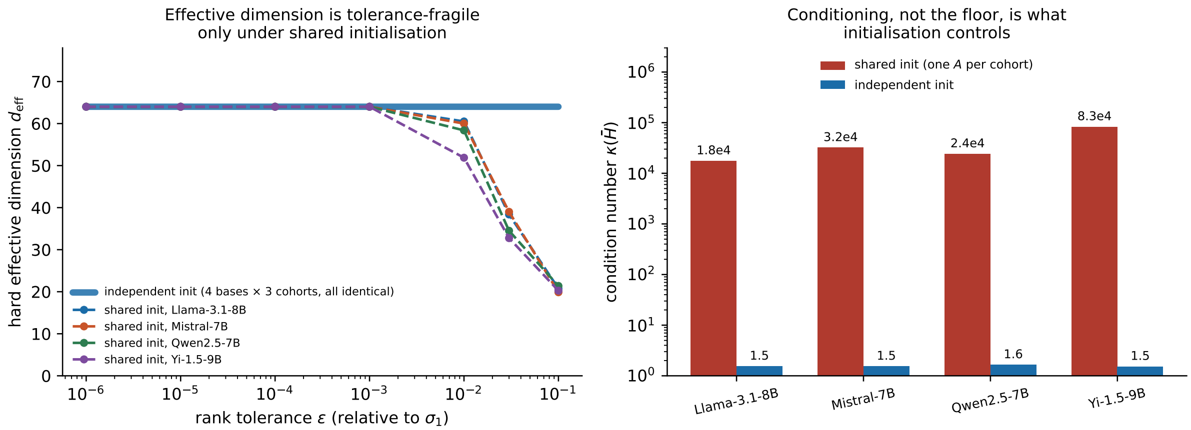

Fig. 1. The two regimes. **Left:** hard effective dimension against rank tolerance. All twelve independently initialised cells hold $d_{\text{eff}} = Tr = 64$ at every tolerance across four decades; the shared cohorts match only at numerically loose tolerances and fall to ≈ 20 by $\epsilon = 10^{-1}$. Since $d_{\text{eff}} = Tr$ holds exactly in both, the irreducible floor is zero in both (§4.3). **Right:** what does separate them, the condition number of $\bar{H}$, four to five orders of magnitude apart.

and not a choice of metric. What is *not* surrogate-independent is the specific value of κ, which would change under a Fisher-weighted H_t with a spread spectrum. The authors therefore read $\kappa(\bar{H})$ as a calibrated index of subspace overlap rather than as the condition number any particular solver will encounter, and they have not measured the latter.

A mismatch of objects, and what closes it. Three different objects appear in the argument so far, and the authors flag that rather than let a reader find it. Theorem 4 constrains $w^\star \in \mathbb{R}^d$ with no rank constraint. The interpolator $\bar{\tau}_H$ whose norm the authors report above has rank up to $Tr = 64$. But the authors' merge pipeline truncates every method back to rank $r = 16$ for deployment parity, so the object whose norm they measure is not one the pipeline can return, and the ridge sweep of §5.2 tests conditioning through a rank-16-constrained solver.

That mismatch is testable, so the authors tested it: they repeated the ridge sweep with the post-merge truncation disabled (§5.6), and it does survive: the unregularised shared-versus-independent ratio is 25 to 117 times, against 2.1 to 65 with truncation, on all four base models. Truncation was not producing the effect. It was *masking* part of it, which is what one should expect, since discarding all but the leading 16 directions of a rank-64 interpolator discards most of the ill-conditioned subspace along with them.

What remains open is only the first of the three objects. The theorem constrains an unconstrained $w^\star$, and every merge the authors run is rank constrained, so the bound bears on their measurements by analogy at that step and not by derivation. That is a real limitation and it is not closed by anything in this paper.

4.5 The audit

Everything above is computed from the adapter weight matrices: no forward passes, no evaluation data, no merging. It costs about one CPU minute for a cohort of four rank-16 adapters. What it reports is not "how much of your loss is irreducible", which is zero in every case the authors have seen, but where your cohort sits on the conditioning axis of Table 2, which §5.2 shows determines whether an unregularised solve-based merge will work at all.

Algorithm 1 The pre-merge conditioning audit. It reads only the A factors of the adapter files: no B factors, no base weights, no forward passes and no evaluation data, which is what makes it runnable before committing to a merge.

Require: Adapters $\{(A_t, B_t)\}_{t=1}^{T}$ trained on a common base; layer sample $\mathcal{L}$; thresholds $c_{\mathrm{hi}} = 0.9$, $c_{\mathrm{lo}} = 0.5$, registered in advance (§F.4) and used unchanged
Ensure: Median principal-angle cosine $\bar{c}$, and a regime label

$C \leftarrow \emptyset$
for each layer $\ell \in \mathcal{L}$ **do**
  for $t = 1$ **to** T **do**
    $V_t \leftarrow \mathrm{orth}\big(A_t^{(\ell)\top}\big)$ ▷ orthonormal basis of task t's row space, r columns
  end for
  for each pair $i < j$ **do**
    $C \leftarrow C \cup \mathrm{svals}\big(V_i^\top V_j\big)$ ▷ cosines of the principal angles
  end for
end for
$\bar{c} \leftarrow \mathrm{median}(C)$ ▷ median over pairs and over layers
if $\bar{c} > c_{\mathrm{hi}}$ **then**
  return ($\bar{c}$, COLLAPSED)
else if $\bar{c} < c_{\mathrm{lo}}$ **then**
  return ($\bar{c}$, SEPARATED)
else
  return ($\bar{c}$, INDETERMINATE)
end if

The whole of it is five lines, stated in full as Algorithm 1. Given T adapters, for each of a sample of layers, orthonormalise the rows of each adapter's A factor and measure how aligned the resulting subspaces are.

How to read what Algorithm 1 returns:

- $\bar{c} > 0.9$ — **collapsed**. Do not run an unregularised solve-based merge. Add a ridge, starting at $\lambda \approx 0.01$, or use a method that averages.
- $\bar{c} < 0.5$ — **separated**. Conditioning is not your problem; the ridge buys between 0.00 and 0.10 nats.
- Between the two — one of the authors' 45 public cohorts landed here and they have no advice for it.

Cost $O(T^2 r^2 d)$ per layer: about one CPU minute at $T = 4$, $r = 16$ over eight layers, with no forward passes and no evaluation data. Implementation: `measure_subspace_geometry.py`.

The 0.9 and 0.5 cutoffs are those registered before the public-cohort audit of §4.6 and are used unchanged here. They are not tuned, and on this evidence they do not need to be: the two populations differ by more than an order of magnitude in $\bar{c}$, so any cutoff between them gives the same partition, and 44 of 45 public cohorts fall outside the interval entirely. **The $\lambda \approx 0.01$ is a starting point, not a derived optimum.** §5.2 finds $\lambda^\star = 0.01$ on seven of eight arms, but also finds that how much regularisation is needed does not track the conditioning, so a short sweep is worth more than the authors' number.

Can I repair the adapters instead of the merge? It is the first thing a practitioner with a collapsed cohort and no budget to retrain will think of, so it deserves an answer even though the authors did not test it. Orthogonalise or re-randomise the A factors, the thought goes, and the subspaces separate.

Table 3. Public LoRA cohorts under the audit of §4.5. The sampling frame and the 0.9 threshold were registered before any adapter was downloaded. The two populations are separated by more than an order of magnitude in median principal cosine and do not overlap: only one cohort of 45 falls between 0.5 and 0.9.

	cohorts	median principal cosine	$\|A_i - A_j\|_F / \|A_i\|_F$
collapsed, cos > 0.9	17	0.915 – 1.000 (med. 0.996)	0.017 – 0.760
intermediate, [0.5, 0.9]	1	0.868	—
separated, cos < 0.5	27	0.049 – 0.440 (med. 0.076)	med. 1.414

The trouble is that A cannot be changed on its own. What the model computes is BA, so replacing A_t by QA_t for any invertible Q leaves the adapter's function unchanged only if B_t becomes B_tQ^{-1}, and then the product is identical and so is every quantity in this section: the row space $\text{range}(A_t^\top)$ is unchanged by an invertible map applied on the left, so the principal cosines, $\bar{H}$ and its conditioning are all exactly what they were. A repair that preserves each adapter's behaviour cannot change the geometry, because the geometry is a property of that behaviour. A repair that changes the geometry changes what each adapter does, and then the merge is being asked to combine different models from the ones that were validated.

There is a narrow exception worth naming: the row spaces here have dimension $r = 16$ inside $d = 4096$, and the near-collinearity is between spaces, not within them, so one could project each A_t onto a private slice of a partition of the ambient space. That is a real construction and it is not free: it discards the components of each task's update outside its assigned slice, which is a lossy edit to a trained adapter, and whether the loss is tolerable is exactly the question the merge was supposed to answer. The authors have not measured it, they do not recommend it on the strength of an argument, and they flag it as the natural next experiment for anyone who cannot retrain.

What the authors do recommend for a cohort that is already collapsed is the ridge, which §5.2 and §5.8 measure, costs one hyperparameter, and leaves the adapters untouched.

4.6 The collapsed regime occurs in public adapters

Every cohort measured so far is one the authors trained themselves, in both conditions, deliberately. That is a controlled experiment and it establishes nothing about anyone else's adapters. So the authors ran the audit of §4.5 over public LoRA cohorts, under a protocol registered before any adapter was downloaded (§F.4). The registration exists because the failure mode here is selection: an author who picks which cohorts to measure can produce any prevalence they like.

Sampling frame, fixed in advance. The authors walk the HuggingFace model index filtered to `library_name=peft`, sorted by downloads descending, and group adapters into cohorts by (publisher, base model). Sorting by downloads rather than recency or search relevance is a choice made in advance and stated: it weights the sample towards adapters people actually use. A cohort enters if it has at least two adapters on one base model, each a readable PEFT LoRA of rank ≥ 4 sharing at least one target module. Every criterion is checkable from metadata before any weight is fetched, and **no cohort was skipped after inspection**. The authors walked 1000 models and stopped at the registered target of 50 cohorts; 330 candidates were rejected as cohorts of one and 282 for having no readable adapter configuration. Five of the 50 could not be scored, leaving 45.

Result. Seventeen of the 45 cohorts, 37.8%, have a median principal cosine above 0.9: the collapsed regime, in adapters the authors did not train and did not select. The registered prediction was 20%.

The distribution is bimodal, and one number checks the measurement. Table 3 is not a continuum with a threshold drawn through it. Forty-four of 45 cohorts sit at one extreme or the other, and the gap between the populations is wider than the spread within either.

The second column is an unplanned validation. If two adapters draw A independently and A moves little in training, then $\|A_i - A_j\|_F^2 \approx \|A_i\|_F^2 + \|A_j\|_F^2$, so the ratio should be $\sqrt{2} \approx 1.414$. The separated cohorts return a median of exactly 1.414. The collapsed cohorts run from 0.017, adapters whose A factors agree to within 1.7%, up to 0.760. The authors did not design this as a check and it is not part of any registered rule, but it is the strongest evidence they have that the audit measures what they claim: the separated population behaves numerically like independent draws, because that is what it is.

What this does and does not establish. It establishes that the regime is common in released adapters, at a rate the authors can state with a frame a reader can rerun. It does not establish that anyone has merged these particular cohorts, or that a published merging benchmark is affected, and the authors make neither claim. Two of the 45 have members at differing LoRA rank, so their $\|A_i - A_j\|_F$ comparison is undefined and only their cosine is reported; both are separated. The audit's cost is unchanged from §4.5: about one CPU minute per cohort, and the entire sweep of 45 cohorts needed no GPU.

Two of the three registered parts. The pre-registration for this audit (§F.4) has three parts. The prevalence figures above are Part C. Part A, the library condition, is reported below. **Part B has not been run**, so the authors do not treat the precondition as fully discharged.

Part A: when does a library hand two adapters the same A? The registration asks for the exact condition, from the source at a pinned version, and fixes no threshold: it is a factual question about a library. The version is not the authors' to choose, since every adapter they shipped records the library that wrote it, and all of them record PEFT `0.19.1`. In `src/peft/tuners/lora/layer.py`, `reset_lora_parameters` initialises A with `kaiming_uniform_` and B with zeros whenever `init_lora_weights` is true, which is the default and what all the authors' configs record. `kaiming_uniform_` draws from the *global* torch RNG; PEFT takes no per-adapter seed and does not reseed around the call. Two separate invocations therefore receive identical A exactly when the global RNG is in the same state as each layer is built, which needs the same seed set beforehand, the same modules targeted in the same order at the same rank, and the same device and dtype. Nothing about the task or the data enters, so the condition is settled before the first gradient step. That is why one seed line in a launcher decides it for a whole cohort, and why it is so rarely reported.

The authors' own cohorts meet the condition by construction in the shared arm, where all four task configs carry one seed, and violate it by construction in the independent arm, where the four seeds differ. Both are checkable from the configs. What cannot be checked from the shipped adapters is the initial A itself: those tensors are post-training, and training moves A, so bitwise identity is not the right instrument and Appendix D reports the residue instead.

The distinction between the parts matters for what may be claimed. Part C establishes prevalence in released adapters, which is what §1 and the abstract assert and all they assert. Part B is what would license a statement about published merging *benchmarks*, and the authors make none: the benchmark-confounding claim is withdrawn in §5.7 and stays withdrawn.

4.7 Scope: one adapter in four did not learn

On all four bases and in both conditions, the translation adapter fails to beat the base model at its own task or is beaten there by another task's adapter; the other three tasks train cleanly, improving 10 to 70% over base with the specialist best on its own task in every cell (Appendix D). The authors' cohorts are therefore $T = 4$ nominal and 3 effective, and they describe them that way throughout.

The authors know why, and it is a data-pipeline defect rather than a training failure. Failing on four bases in both arms is too systematic to be a bad run, so they diagnosed it, which needs no retraining. Translation is the only task in the matrix configured to stream its training set, and the streaming branch of the authors' loader takes the first n examples in corpus order, while the branch every other task takes shuffles first. WMT19 de-en is a concatenation of sub-corpora of roughly 35M pairs, so an unshuffled prefix of 7500 samples whichever sub-corpus is written first rather than the corpus. Measuring the prefix the trainer actually saw against a shuffled sample of the same split and against the evaluation split settles what happened: the prefix is parliamentary proceedings, at 17.4 occurrences per 1000 tokens of a small set of parliamentary terms against 0.6 in the shuffled control and 2.8 in the news-domain evaluation split, with a type-token ratio of 0.11 against 0.30 shuffled. The adapter was trained on 7500 consecutive sentences of one narrow register and evaluated on another.

That is a defect in the authors' configuration, not a property of translation as a task or of LoRA. **The authors report it rather than repairing it**, because repairing it means retraining the cohort and re-running every cell measured on it, and the pre-registered comparisons in §5 are all paired *within* a cohort: a cohort where one member is weak is a valid cohort for asking whether initialisation geometry changes a merge, so long as both arms contain the same weak member, which they do. What it is not is a cohort of four well-trained adapters, and the authors make no claim about that case.

Two further things bound the damage and one does not. Worst-task excess is never driven by the degenerate adapter: across the 84 independent-cohort cells the argmax is `gsm8k` in 64% and `alpaca` in 36%, and translation never. And every geometry and conditioning quantity above is reported at both $T = 4$ and $T = 3$ in Appendix D; the shared-versus-independent contrast is unchanged to three decimals. What is *not* bounded is that the merged models contain an untrained member. That is arguably realistic, since practitioners merge adapters of uneven quality, but it is not the same experiment as four well-trained adapters, and the authors make no claim about the latter.

5 Pre-Registered Tests of What This Predicts

The previous section leaves every cohort in the floor-zero regime, so all observed worst-task excess is in principle recoverable, and separates the cohorts on conditioning instead. Both halves of that invite a test. The first invites an encoder that recovers the slack, and the authors built one: a rate-distortion encoder using incoherent orthogonal mixing and scalar quantisation, regularised with a single Tikhonov ridge term (Appendix A.3, implementation in Appendix G). On a shared-initialisation cohort it wins comfortably against every baseline, which is what made the test below necessary. The second invites the question of whether a four-order-of-magnitude difference in conditioning is visible to anything a practitioner does.

This section reports what happened when the authors tested that result properly. Every decision rule quoted below was committed to version control before the corresponding compute ran. Appendix F summarises the rules and gives the commit ordering that makes them checkable.

5.1 The encoder's advantage does not survive

Pre-registered question. Does the encoder's win over the best baseline survive the move from the shared-initialisation cohort to independently initialised ones? The rule, fixed in advance: for each base model, compare the encoder against the champion baseline selected on the mean over the three independent cohorts, with a tie threshold of 0.005 nats and a one-directional noise gate at $2 \times \mathrm{SE}$ that may only downgrade a win or loss to a tie, never promote.

Outcome: it does not survive. The registered arm gives one win, one tie and two losses, and each control removes another piece: matching the inference path turns Mistral's loss into a tie, and dropping the untrained adapter turns Llama's win into a tie, leaving **no wins on any base**. Both pre-specified robustness lines agree

Table 4. Pre-registered replication on independently initialised cohorts, and the two controls applied to it. Entries are the paired difference d in worst-task NLL excess (nats), mean over three cohorts; positive favours the encoder. *As published* is the registered primary arm. *Matched* corrects an inference-path mismatch: at d_{eff} realisation the merged adapter has rank 64 against the baselines' 16, and the fused fast path does not take mixed-rank adapters, so the encoder alone ran vanilla TRANSFORMERS; the two paths differ by up to 0.0105 nats on identical inputs. The rank-16 realisation is matched on both counts and needs no new measurement. $T = 3$ additionally drops the adapter that never learned its task (§4.7). Levels are not comparable across T, so only the verdict transfers. The noise gate ($2 \times$ SE, one-directional) downgraded nothing; SE on d runs 0.0002 to 0.0012.

base	as published	matched	matched, $T = 3$	verdict
Llama-3.1-8B	+0.0335	+0.0257	+0.0033	wins → **ties**
Mistral-7B	−0.0055	−0.0022	−0.0012	loses → **ties**
Qwen2.5-7B	−0.0002	−0.0004	+0.0009	ties throughout
Yi-1.5-9B	−0.0150	−0.0154	−0.0053	loses throughout
count				1/1/2 → 1/2/1 → **0/3/1**

with the primary arm. Re-selecting the champion inside each cohort, the variant deliberately biased against the authors' method, returns the same one/one/two; applying the original single-cohort rule three times succeeds on `indep1` and fails on `indep2` and `indep3`, so the earlier optimistic reading was the best of three draws. Two further details: the rank-64 realisation disagrees at $T = 3$, keeping Llama at +0.0081, and Mistral's champion changes from TIES to TVQ.

The registered $T = 4$ arm stays primary. The direction is consistent across all three columns, and the reading the authors take is that the encoder's advantage was substantially a property of the degenerate cohort it was developed on. On that cohort $\bar{H}$ is ill-conditioned by four orders of magnitude (§4.4), which is exactly what the ridge term was introduced to handle, so the method was tuned on the one regime where its distinguishing component has work to do. §5.2 makes that concrete: the ridge gain is 0.19 to 9.73 nats on the shared cohort and 0.00 to 0.10 on the independent one. What does not survive is the claim that the encoder is the better method.

5.2 Does the conditioning matter? Only for methods that solve

§4.4 found a four-order-of-magnitude difference in $\kappa(\bar{H})$ between the two initialisation regimes. A difference that size which nothing observable depends on would be a curiosity, so the authors tested it, pre-registering the prediction before running the cells.

First, the null, and what it can actually detect. The authors compare every method on matched cohorts, same method, same base, shared against independent initialisation, with three cohorts per arm so the pre-registered gate can be applied: a cell counts as an effect only if the mean difference exceeds $\max(0.005, 2\,\text{SE})$. Across 20 base-by-method cells, 15 show no detectable effect, 5 favour independent initialisation, and none favours shared.

The number that matters alongside that is the smallest effect the design could have found, because a null without one is not evidence of absence. The gate sits at 0.0052 nats in the median cell and 0.0109 in the worst, so the honest statement is *no effect detectable at about* 0.01 *nats*, not *no effect*. It also disposes of a smaller number that has been quoted in this setting, a mean difference of 0.0014 nats against a metric resolution of about 0.001: that difference sits below the median gate, so it is a quantity this design cannot resolve in either direction.

The five cells that do clear the gate are worth naming rather than averaging away, because four of them are not spread at random. Three are the subspace-alignment method, on Mistral, Qwen and Yi, at 0.0096 to 0.0129 nats; the others are TIES on Llama at 0.0151 and TVQ on Yi at 0.0087. Task arithmetic and DARE, the two methods

Table 5. The heuristics null, measured in downstream accuracy rather than in NLL. Each entry is $d = \mathrm{acc}_{\mathrm{indep}} - \mathrm{acc}_{\mathrm{shared}}$ for one method on one base; a cell counts as an effect only if $|d|$ exceeds both 0.05 and $2\,\mathrm{SE}_{\mathrm{binom}}$, both fixed in advance. No cell does. **This design has one cohort per arm**, so its gate is binomial and its minimum detectable effect is about 0.06 on GSM8K and 0.11 on HumanEval: a null here means no effect detectable at that size, not no effect.

		GSM8K		HumanEval	
base	method	shared	d	shared	d
Llama-3.1-8B	task arithmetic	0.792	−0.016	0.646	−0.006
	TIES	0.762	−0.016	0.659	−0.104
	DARE	0.786	−0.010	0.640	−0.006
	TVQ_2	0.746	+0.002	0.549	+0.030
	KnOTS	0.778	+0.000	0.610	+0.012
Mistral-7B	task arithmetic	0.468	+0.002	0.360	−0.006
	TIES	0.464	−0.010	0.232	+0.018
	DARE	0.484	−0.006	0.378	−0.018
	TVQ_2	0.454	+0.024	0.195	+0.067
	KnOTS	0.496	+0.010	0.287	−0.018
Qwen2.5-7B	task arithmetic	0.514	−0.010	0.744	−0.049
	TIES	0.622	+0.014	0.738	−0.024
	DARE	0.518	−0.022	0.732	−0.006
	TVQ_2	0.616	+0.010	0.768	−0.018
	KnOTS	0.602	−0.018	0.683	+0.055
Yi-1.5-9B	task arithmetic	0.798	−0.008	0.128	−0.024
	TIES	0.786	−0.008	0.171	+0.000
	DARE	0.804	−0.010	0.128	−0.012
	TVQ_2	0.780	+0.002	0.152	−0.030
	KnOTS	0.810	−0.016	0.085	−0.012

closest to a plain mean, show nothing anywhere. That ordering is the one the mechanism below predicts: a method that explicitly operates on the subspace geometry is the heuristic most exposed to a change in that geometry, and it gains from the cohort whose subspaces are not near-collinear. The effect is small, consistent in direction, and the authors do not build anything on it.

The same null in a unit the authors do not disclaim. Everything above is worst-task NLL excess, and §6 reports that this metric does not track downstream accuracy across methods and on one benchmark ranks them backwards. That objection lands harder on a null than on an effect: the differences here are hundredths of a nat, which is the range in which the authors have told the reader not to trust the metric. Meanwhile the positive result beside it, the solver collapse, has two units. Reporting one half in a disputed unit and the other half in two is an asymmetry the authors would rather remove than defend, so they pre-registered the null in accuracy as well and ran it.

Zero of the forty cells show a detectable effect, so both registered predictions hold and by the registered rule the null carries over to accuracy (Table 5). The median $|d|$ is 0.012 and the largest is 0.104. All 80 cells produced a scorable generation for every item, with no empty outputs anywhere, which matters because the scorer defects of §5.8 were what made an earlier version of these benchmarks unusable; both arms were re-run here rather than reusing the existing shared-arm cells, precisely so that no comparison straddles a scorer change.

Table 6. What each solution does with the ill-conditioned directions. Amplification is $\|\Delta_{\text{merged}}\|/\|\bar{\Delta}\|$; tail fraction is the share of squared norm in the bottom quartile of $\bar{H}$'s spectrum. Ranges are over the four base models. On the independent arm a tail fraction near 0.21 is what a generic direction gives when a quarter of the dimensions holds a quarter of the energy, so the signature is the 0.73 to 0.80 on the shared arm, not the baseline it is measured against.

	shared		independent	
solution	amplification	tail	amplification	tail
mean task delta	1.00	0.000	1.00	0.21
task arithmetic	0.99	0.000	0.86–0.92	0.20
DARE	0.98–0.99	0.000	0.86–0.90	0.20
TIES	1.98–2.04	0.000	2.04–2.22	0.21
TVQ (b=2)	1.95–2.16	0.000	1.68–1.95	0.20
RegMean, minimal ridge	35.4–51.4	0.73–0.79	3.41–3.67	0.29
the authors' encoder, $\lambda = 0$	36.7–56.7	0.78–0.80	3.42–3.67	0.29
their encoder, $\lambda^\star$	2.20–2.56	0.000	2.84–3.05	0.28
exact interpolator $\bar{\tau}_H$	43.4–64.1	0.62–0.66	3.96–4.00	0.28

The strength of that null is bounded by the design and the authors state the bound as registered: with one cohort per arm the gate is binomial, so this is **no effect detectable at about 6 accuracy points on GSM8K and 11 on HumanEval**, not no effect. A real effect of three or four points would sit inside it undetected.

Two things in the table are worth a reader's attention and neither is settled by the registered rule.

One cell nearly cleared, in the direction that would have hurt. TIES on Llama at HumanEval gives $d = -0.104$ against a gate of 0.107: 97% of the way, and negative, meaning the *shared* cohort scored higher. Had it cleared it would have been the one outcome the registration named as most informative against the authors. It did not clear and the authors do not treat it as an effect, but a reader recomputing this table will see it and should see it here first.

The signs lean. Of the 38 non-zero cells, 26 favour the shared arm and 12 the independent one, which an exact sign test puts at $p = 0.034$, with the lean present on both benchmarks. **The authors did not register a sign test**, so this is an observation and not a verdict, and they neither claim an effect from it nor let it modify the registered outcome. The authors report it because it points the opposite way from the NLL null, whose five clearing cells all favoured independent initialisation. Taken at face value the two units disagree about a direction while agreeing that nothing is large. The honest reading is that both are consistent with no real difference plus noise, and that a design able to resolve a few accuracy points, which means several cohorts per arm rather than one, is what would settle it. That is a genuinely new experiment and it needs its own registration.

Then the mechanism, measured. The explanation for the other 15 cells is that those methods never go near the ill-conditioned direction. That is a measurement rather than an assertion, and it needs no forward passes: every merge here is weight arithmetic on the adapter factors. For each merged solution the authors report its norm relative to the mean task delta, and the share of its energy lying in the bottom quartile of $\bar{H}$'s eigen-directions, in the projected coordinates of §4.4.

Table 6 gives it. On the ill-conditioned cohort every heuristic sits at one to two times the mean task delta with a tail fraction that rounds to zero at three decimals, while both unregularised solvers sit at 35 to 57 times with roughly three quarters of their energy in the bottom quartile of the spectrum. They are the only methods that go there, which is why they are the only methods the conditioning reaches. One incidental check falls out: RegMean

Table 7. Worst-task NLL excess for the unregularised solve ($\lambda = 0$), and the gain from the best ridge. Conditioning is invisible to the heuristics above and severe here.

base	$\lambda = 0$			ridge gain $L(0) - L(\lambda^\star)$	
	shared	independent	ratio	shared	independent
Llama-3.1-8B	0.3081	0.0713	4.3×	0.2216	0.0000
Mistral-7B	9.7799	0.1504	65×	9.7301	0.0952
Qwen2.5-7B	0.1973	0.0256	7.7×	0.1867	0.0114
Yi-1.5-9B	0.2490	0.1209	2.1×	0.2211	0.0904

at minimal ridge and the authors' encoder at $\lambda = 0$ agree to two decimals on the independent arm across all four bases, so two differently derived solvers land on the same solution once the problem is well posed.

So the authors sweep the ridge, on two solvers. To expose the conditioning one needs a method that actually solves the system. The authors' does, and it carries a Tikhonov ridge, so sweeping λ over seven values, 0, 0.01, 0.03, 0.05, 0.13, 0.30 and 1.00, on both cohorts with rank pinned to 16 in every cell isolates the effect.

Table 7 gives the result. Unregularised, the same merge is 2.1 to 65 times worse on the ill-conditioned cohort; on Mistral it collapses outright to 9.78 nats against 0.15. The ridge gain is larger on the shared arm on all four bases, which was the pre-registered prediction and held.

The other half of the prediction failed. The authors also predicted that $\lambda^\star$ itself would be at least ten times larger on the shared arm. It is not: $\lambda^\star$ is *identical* across arms on Mistral, Qwen and Yi (0.13, 0.13, 0.30), separating only on Llama. How much regularisation is needed does not track conditioning in any simple way; how much it buys does. By the pre-registered rules this makes the test PARTIAL, and the authors claim only the half that held.

5.3 Both arms at three cohorts

Table 7 rests on one shared cohort against three independent ones, so its noise gate borrowed its standard error entirely from the independent side. A shared-versus-independent contrast measured that way is three draws against one, and a reader is right to discount it. The authors registered the completion in advance and ran the sweep on two further shared cohorts, so both arms are $n = 3$ and the gate carries a term from each. This is a replication of a result the authors had already published, so it could only cost them something; the registration says so and fixes the consequence text for the outcome where it fails.

It does not fail. With $\mathrm{SE} = \sqrt{\mathrm{SE}_s^2 + \mathrm{SE}_i^2}$ and the gate at $\max(0.005, 2\,\mathrm{SE})$, the ridge gain is larger on the shared arm on **four of four** bases and the unregularised excess is worse there by at least a factor of two on **four of four**: ratios of 4.42, 76.2, 10.8 and 2.82, against gates of 0.024, 4.26, 0.128 and 0.166. Mistral's shared arm averages 11.72 nats at $\lambda = 0$ against 0.154 independent.

One quantity is worth reporting because it could have embarrassed the earlier table and does not. The published single-cohort value, `seed1`, sits 0.23, 0.53, 0.75 and 0.66 standard deviations from its own arm's three-cohort mean. The authors registered that a distance above 2 on two or more bases would oblige them to say the published value was unrepresentative of its own arm. It is representative, and Table 7's numbers stand as single-cohort estimates of quantities the three-cohort means confirm.

Table 8. RegMean on both cohorts. Its minimally regularised reference is $\lambda = 10^{-6}$ rather than 0: RegMean solves in the full input dimension against a Gram of rank at most $Tr = 64$ against $d = 4096$, so $\lambda = 0$ is singular rather than ill-conditioned and returns null-space noise without raising an error. The authors' encoder solves inside range($\bar{H}$), so its $\lambda = 0$ is defined. The two columns are therefore not the same object and each method is compared only against itself across arms.

base	cohort	$\lambda = 10^{-6}$	$\lambda^\star$	gain	ratio
Llama-3.1-8B	shared	1.9789	0.01	1.7428	27.9×
	independent	0.0708	10^{-6}	0.0000	
Mistral-7B	shared	10.6283	0.01	10.5314	70.9×
	independent	0.1499	0.01	0.0449	
Qwen2.5-7B	shared	0.2376	0.01	0.2255	9.3×
	independent	0.0256	0.01	0.0133	
Yi-1.5-9B	shared	0.5700	0.01	0.5477	4.7×
	independent	0.1203	0.01	0.0913	

5.4 The same test on a solver the authors did not build

Everything above concerns one solver, and the authors built it. A referee is entitled to ask whether the effect is a property of the family or of the authors' implementation, and the question is sharp enough to be worth answering with a pre-registered experiment rather than an argument.

RegMean (Jin et al. 2023), in the data-free adapter-only form, is a solver the authors did not design. The authors swept it on the same grid and the same two cohorts, registering two predictions in advance: that the ridge gain would be larger on the shared arm, and that the minimally regularised excess would be worse there by at least a factor of two, each on at least three of four bases.

One thing to be exact about, since the authors' practical recommendation reduces to regularising a solve-based merge and RegMean ships a regulariser of its own. Its published regulariser is *not* the Tikhonov ridge swept below: RegMean scales the non-diagonal entries of each Gram by a scalar α, default 0.9, which is a data-set diagonal shrinkage rather than a free parameter. An earlier version of this section described the swept ridge as part of RegMean's published formulation. That was wrong: the λI is the authors', and the comparison below is between their regulariser and theirs. §5.5 reports what the published default does on these cohorts.

Both predictions hold on four of four bases. The effect is not a property of the authors' encoder.

Three details in Table 8 are worth reporting rather than smoothing. $\lambda^\star$ is 0.01 on seven of the eight arms, so on RegMean too the amount of regularisation needed does not track the conditioning even though the amount it buys does; that is the same split the authors' own sweep found, now reproduced on a method they did not write. Llama's independent arm selects the minimal λ and gains exactly zero, which is the cleanest statement of the claim available: on a well-conditioned cohort there is nothing to repair. And from $\lambda = 0.13$ upward the two arms agree to three decimals, because heavy regularisation washes the difference out entirely.

5.5 Does RegMean's published default already fix this?

The authors' practical recommendation reduces to regularising a solve-based merge, and RegMean ships a regulariser. If its default were already strong enough on a collapsed cohort, the recommendation would shrink to the much smaller claim that the audit tells you when the default matters. The question is worth answering rather than leaving for a referee, and it is a lookup plus a CPU-only computation.

RegMean's published regulariser scales the non-diagonal entries of each Gram by α, default 0.9, which is equivalent to adding $(1-\alpha)\,\mathrm{diag}(G)$: a shrinkage whose size the data sets, rather than a free parameter. It does regularise, and on a rank-deficient Gram it does make the system solvable. The question is whether it is enough.

On the shared-initialisation cohorts it is not, by about an order of magnitude. Averaged over sampled layers, $\alpha = 0.9$ is equivalent to an isotropic ridge of 5.2×10^{-4} to 6.1×10^{-4} across the four bases, against the $\lambda^\star = 0.01$ the sweep selects on seven of eight arms: roughly 20 times smaller. The conditioning it leaves behind reflects that. The α-shrunk system has condition number 6.6×10^3 to 9.1×10^3, while $\lambda = 0.01$ brings the same layers to about 1.5×10^2, a factor of 55 apart. The default is on the wrong side of the gap this paper is about.

The independent arm is the control, and it behaves as the story requires. The *same* default leaves κ at 1.2×10^3 to 1.4×10^3 there, six to seven times lower than on the shared arm, while the equivalent ridge it applies is identical to three significant figures (5.2 to 6.1×10^{-4} on both arms). That is the point in one line: α-shrinkage is a fixed dose set by the data's scale and not by the cohort's conditioning, so it neither knows nor cares which regime it is in. On a well-conditioned cohort the dose is ample. On a collapsed one it is 20 times short, and nothing in the method notices.

Two limits on what this shows, both worth stating. It is a statement about conditioning, computed from the adapter factors, not a merge the authors evaluated: they did not run a λ-free α-shrunk RegMean through the NLL harness, so they do not report a task-loss number for it. And α is a documented knob with the default merely being 0.9, so a practitioner who lowers it is doing exactly what §5.2 recommends. The claim here is only that the shipped default does not do it for them.

5.6 The same test without rank truncation

§4.4 notes that the authors' pipeline truncates every merge back to rank r, so the sweep above tests conditioning through a rank-constrained solver while the interpolator whose norm they report has rank up to Tr. Repeating the sweep with truncation disabled removes that mismatch, and the truncated sweep is then the comparison arm rather than being re-run.

Table 9. The conditioning link with rank truncation disabled. Worst-task NLL excess in nats at $\lambda = 0$, and the shared-to-independent ratio with and without truncation. The truncated column is the sweep of Table 7, reproduced here as the comparison arm rather than re-run. **Two entries here invite a second look and the authors have taken it.** Mistral's shared value, 11.6283, sits exactly 1.0000 above RegMean's 10.6283 in Table 8 at four decimals; the underlying numbers are 11.628281 and 10.628309, so the gap is 0.999973 and the agreement is a coincidence between two different methods on two different runs rather than a transcription. And Qwen's independent value matches Table 7's to four decimals, 0.025571 against 0.025638; that one is not coincidence but the point of the table, since on a well-conditioned cohort there is almost nothing in the discarded directions for truncation to remove.

	$\lambda = 0$ excess		shared / independent	
base	shared	independent	untruncated	truncated
Llama-3.1-8B	7.4266	0.0635	116.9×	4.3×
Mistral-7B	11.6283	0.1582	73.5×	65×
Qwen2.5-7B	0.8993	0.0256	35.2×	7.7×
Yi-1.5-9B	3.1294	0.1227	25.5×	2.1×

Table 9 shows the link surviving on all four bases, with the effect *larger* without truncation, by factors of 25 to 117 against 2.1 to 65. Truncation was not manufacturing the result; it was hiding part of it, which is what one should expect when all but the leading 16 directions of a rank-64 interpolator are discarded and the ill-conditioned subspace goes with them. The numbers the authors report as primary are therefore the conservative ones.

What this licenses, and what it does not. It licenses a conditional recommendation: if you merge with a method that solves a linear system, check your cohort's conditioning and regularise, because unregularised it can fail by a factor of 65 at rank 16 and 117 without truncation, on two solvers, one of which the authors did not build. It does *not* license the general advice to initialise independently. On the methods most practitioners actually use the null above stands, with the single qualification that the subspace-alignment method gains a little from independent initialisation on three of four bases, and the authors do not dress that up as anything more.

5.7 A claim the authors withdraw

The result in §5.1 raises an obvious and much more interesting possibility: if method rankings shift when initialisation changes, then published merging benchmarks, which as far as the authors can tell mostly use shared-seed cohorts, might be measuring initialisation geometry rather than method quality. The authors described this on first sight as the strongest result available to them.

The authors pre-registered a control to test it, and the control did not support the claim. The control asks whether the method ranking is stable *within* the independent regime, across the three cohorts that differ only in initialisation draw. If rankings are unstable within a single regime, then ranking changes across regimes cannot be attributed to the regime. Two of four base models had a non-unanimous top-1 method across the three cohorts, above the threshold fixed in advance, so by the authors' own rule the attribution fails and **the claim that published merging benchmarks are confounded by shared-initialisation geometry is withdrawn**.

The authors record one caveat, and they deliberately did not let it reverse the verdict. The control as written counts a top-1 change as instability regardless of margin, and both flips involve methods separated by far less than the tie threshold: on Qwen the top two are 0.0140 and 0.0141, a coin flip. A margin-aware version of the control would very likely give a different answer. But that version was not what the authors pre-registered, and rewriting a decision rule after seeing which way it fell is exactly the practice the pre-registration exists to prevent. The authors stated the defect and left the verdict, and registered a *second* control rather than editing the first.

A second control, reported but not used to reinstate anything. The authors registered a margin-aware version of the rule afterwards, in the same document that governs §5.4: a base counts as unstable only if its top-1 method changes *and* the margin between the top two exceeds the tie threshold in at least one cohort where the change occurs. It needs no new cells. Run as written it returns **zero of four** bases unstable against two of four for the margin-blind rule, because both flips sit well inside the threshold: Mistral's margins are 0.0004, 0.0003 and 0.0011 nats and Qwen's 0.0001, 0.0006 and 0.0003, against 0.005.

Read literally, that rule would reinstate the benchmark-confounding claim as provisional. **The authors decline to reinstate it**, and the reason is worth stating plainly because it cuts against them. This control was registered after its data had been seen. The registration says so about itself, and the authors could bound it in various ways, but no amount of disclosure changes the fact that they would be resurrecting a claim they had already withdrawn using a rule written once they knew how the first one fell. That is the single thing pre-registration exists to prevent, and a paper making this much of the practice should not spend its credibility on a claim it does not need. The prevalence audit already carries the load of showing this matters outside the authors' own repository.

Table 10. Downstream accuracy of the same merges, with the repaired scorers. GSM8K exact match at $n = 500$, HumanEval pass@1 at $n = 164$, its whole test split. Gate is $\max(5 \text{ points}, 2\,\text{SE})$ on the binomial. No cell had a scorer discard: every generation was scored.

	shared			independent		
base	λ=0	$\lambda^\star$	gain	λ=0	$\lambda^\star$	gain
GSM8K, exact match						
Llama-3.1-8B	0.318	0.654	+0.336	0.708	0.712	+0.004
Mistral-7B	**0.000**	0.462	+0.462	0.276	0.424	+0.148
Qwen2.5-7B	0.496	0.808	+0.312	0.786	0.786	+0.000
Yi-1.5-9B	0.346	0.802	+0.456	0.698	0.794	+0.096
HumanEval, pass@1						
Llama-3.1-8B	0.250	0.573	+0.323	0.518	0.530	+0.012
Mistral-7B	**0.000**	0.244	+0.244	0.067	0.189	+0.122
Qwen2.5-7B	0.530	0.744	+0.213	0.726	0.738	+0.012
Yi-1.5-9B	0.024	0.091	+0.067	0.049	0.067	+0.018

The margin-aware numbers stay in the paper as a measurement, because they say something true and useful: the ranking flips that triggered the withdrawal are all inside noise. That is a statement about the fragility of single-draw rankings, not about published benchmarks.

So the claim the authors make is this, and no more: method rankings on this metric move between cohorts that differ only in initialisation draw, the movement is within noise once margin is accounted for, and the collapsed regime that would make such rankings hard to interpret is common in released adapters. The benchmark-confounding claim stays withdrawn. The authors have not measured a published merging benchmark, and they do not claim one is wrong.

What survives independently of all of it is weaker and still worth saying: single-seed method rankings on this metric are not reproducible, so merging comparisons should report multiple initialisation draws.

5.8 The collapse is visible in downstream accuracy

Everything above is measured in worst-task NLL excess, and §6 reports that this metric does not predict downstream accuracy across merge methods, and on HumanEval predicts it backwards. A reader is entitled to extend that objection to the whole paper: if the metric does not track model quality, why should any of these numbers matter?

The objection is about differences of hundredths of a nat between methods. The conditioning effect is not that. At $\lambda = 0$ the shared-versus-independent gap runs from 0.22 to 11.56 nats across the four bases at $n = 3$ per arm (0.13 to 9.63 from the single-cohort Table 7, which is the same quantity measured with one draw), one to three orders of magnitude above both the metric's resolution and the differences the correlation was computed over. An effect that size either shows up in accuracy or the metric is measuring something other than model quality. The authors registered the test without knowing which, and fixed in advance that a null would move this limitation into §1 and delete the practice recommendations rather than defend them.

It shows up, on both benchmarks. The ridge improves the shared arm past the gate on **four of four** bases on both benchmarks. The difference-in-differences, which is the claim that the ridge helps *because of* conditioning rather than helping everything everywhere, clears the gate on four of four on GSM8K and three of four on HumanEval; the exception is Yi, where both arms sit near the floor of a benchmark it barely solves at all (0.024 to

0.091). By the registered rule, requiring either prediction on at least three of four bases on at least one benchmark, the outcome is CONFIRMED.

The number worth stating plainly. On Mistral, the shared-cohort unregularised merge scores **zero** on both benchmarks. It is not producing malformed output that a scorer discards: all 500 GSM8K generations and all 164 HumanEval generations are non-empty, and none contains an extractable answer or passes a test. The model generates fluent text and has lost the capability entirely. One Tikhonov term on the same adapters recovers 0.462 and 0.244. That is what four orders of magnitude of conditioning buys in a unit a practitioner cares about.

What this does not license. It does not show that worst-task NLL excess predicts accuracy in general, and the authors do not claim it. The correlation that fails in §6 is across merge *methods* at fixed cohort, and nothing here tests that. What it shows is that the specific effect this paper is about is not an artifact of the metric, which is the objection it was registered to answer.

5.9 The predicted rate exponent is unresolved

Theorem 4 predicts distortion decaying as $2^{-2R/d_{\text{eff}}}$, and the authors registered that as a slope in $[-2.4, -1.6]$ on at least three of four bases. **The sweep cannot resolve it, and two earlier readings of it are withdrawn.** A previously reported falsification rested on asymptotes computed at a different ridge value, and the refitted sweep turns out to have too little dynamic range to measure an exponent at all: only two of the six quantiser widths sit above the metric's own noise floor, one of which the registered rule excludes as clipping-dominated. By the rule as written the outcome is UNRESOLVED, and this removes the only positive evidence for the rate term anywhere in this paper. Appendix E gives the sweep, both faults and the rule applied to the result.

One operational regularity does survive at the precision available: with the ridge on, every base is within 0.005 nats of its own asymptote from three bits upward, so a practitioner quantising a merged adapter can stop at three or four bits regardless of any exponent.

5.10 DARE: a negative result with a mechanism, and its limits

DARE (Yu et al. 2024) drops a random fraction of task-vector entries and rescales the survivors by $1/(1-p)$. Composed with task arithmetic it does essentially nothing in the authors' setting: across a density sweep from 0.05 to 0.5 on four base models, it beats plain task arithmetic in three of twenty cells and all three are within 0.0005 nats, a fifth of their tie threshold.

Unlike most negative results this one has a mechanism that predicts its own shape. The rescaling makes the masked task vector an unbiased estimator of the unmasked one, $\mathbb{E}[D_{\text{DARE}}] = D_{\text{TA}}$. The authors state the consequence carefully, because the obvious version of the argument is not available here: negative log-likelihood of an LLM is *not* convex in the weights, so Jensen's inequality does not apply globally and they do not claim $\mathbb{E}[L(D_{\text{DARE}})] \geq L(D_{\text{TA}})$ as a theorem. What does hold is the local statement. Expanding L to second order about D_{TA}, the first-order term vanishes by unbiasedness and the penalty is $\frac{1}{2}\operatorname{tr}(H\Sigma) + o(\|\Sigma\|)$ with $\Sigma \propto (1-p)/p$ and H the local Hessian. Where $H \succeq 0$, which is the generic situation near a trained solution, this is non-negative and grows as density falls. That is what the authors observe, on three of four base models. The authors verified their implementation against the reference one and confirmed the operator reproduces the analytic signature $\sqrt{(1-p)/p}$ to three decimals, so this is a property of the method and not of their code.

The negative result does not generalise, and the authors tested that. The Jensen argument applies only to the task-arithmetic composition, since TIES is biased and the mask changes which coordinates survive trimming. The authors pre-registered the DARE-TIES composition test before implementing it. The outcome is mixed: DARE composed with TIES *helps* decisively on Llama (+0.0462 nats, against a gate of 0.0036) and hurts decisively on the other three. By the rule fixed in advance this is a mixed verdict and neither direction is claimed.

So the correct scope for the negative result is DARE composed with task arithmetic, not DARE. The authors flag that Llama is now the odd model out in two separate pre-registered tests, here and in Table 4, without proposing a mechanism for it; a base-level moderator that neither pre-registration anticipated deserves a designed experiment rather than another post-hoc reading.

6 Discussion and Limitations

What the authors think survives. That whether a cohort's adapters shared one initialisation of A determines the conditioning of $\bar{H}$, by four orders of magnitude, measured across four base models and four cohorts, with an amplification returning the orthogonal-subspace value T to within 1.4% at two values of T. That the regime is common in adapters the authors did not train: 17 of 45 public cohorts, under a frame fixed before any download. That the collapse is invisible to merging heuristics and decisive for solve-based merges, pre-registered, holding on a second solver the authors did not design, surviving the removal of rank truncation with the effect larger rather than smaller, and measured in two units, worst-task loss and downstream accuracy, which agree. That the mechanism is visible in the merged weights: the heuristics stay within a factor of two of the mean task delta with essentially none of their energy in the ill-conditioned directions, while the unregularised solvers reach 35 to 57 times with three quarters of theirs there. And that one CPU minute on the adapter files tells a practitioner which case they are in, which is the part of this the authors expect anyone to use.

Two smaller things survive on the theory side: that the bound's floor term is *vacuous* for LoRA cohorts in general position, and the exact instance form $B^2 - \bar{\tau}_H^\top \bar{H} \bar{\tau}_H$ that should be evaluated in its place. The bound and its construction are mathematics and do not depend on any empirical outcome here, but §3 is the authors' account of how little that buys.

What does not, collected in one place. These are scattered through the paper at the point each is established, which is where a reader checking a number needs them. Collected here so that a reader who wants the list does not have to assemble it.

(1) **The authors' own encoder's advantage.** Three controls each removed a piece of it: independent initialisation, matching the rank and inference path to the baselines, and dropping the adapter that never trained. After all three it beats the best baseline on no base model and loses on one. The middle control moved a verdict in the authors' favour and the other two against them (§5.1).

(2) **That published merging benchmarks are confounded by initialisation geometry.** Withdrawn by a control the authors wrote themselves. A later margin-aware control would reinstate it, and the authors decline to use it that way because it was registered after its data had been seen (§5.7).

(3) **That the finite-rate exponent is falsified.** Its two supporting slopes were fitted against an asymptote computed at a different ridge value. Against matched asymptotes the sweep has too little dynamic range to measure the exponent at all, so the question is unresolved rather than settled either way (Appendix E).

(4) **A sixty-fold difference in the irreducible floor.** An artifact of substituting a soft participation ratio into a formula derived for a rank. The floor is zero in both regimes (§4.3).

(5) **That subspace-alignment methods have no overlap to exploit.** Formal independence is not the absence of overlap, and the authors' KnOTS implementation reduced to task arithmetic under its default inner combination, so it never tested the question (§2).

(6) **Worst-task NLL excess as a proxy for deployed capability.** It does not predict downstream accuracy across methods on either benchmark the authors can test, and inverts on one. The scope of that failure is narrower than it looks, and §6's second limitation is where the authors argue so.

The authors have left each of these at full strength rather than retreating to the subset that flatters the method, because the conditioning result is only worth anything if a reader can believe they would have reported it the other way.

Limitations.

(1) **The metric's resolution is about 0.001 nats, and several of the authors' differences are near it.** A zero-mean perturbation changing the merged delta by 37 to 57% in Frobenius norm costs roughly that much. Any comparison in this paper at the 0.005 threshold should be read with that in mind; it is why the authors decline to call two of the four bases in Table 4 losses. The conditioning results are not affected: the shared-versus-independent difference at $\lambda = 0$, which is the quantity §5.2 turns on, runs from 0.13 to 9.63 nats on the single-cohort table and 0.22 to 11.56 at $n = 3$ per arm (§5.3), one to three orders of magnitude above the resolution either way.

(2) **Worst-task NLL excess does not predict downstream accuracy, and on one benchmark it predicts it backwards.** Computed within a single row of five or six methods this is unmeasurable, since $\mathrm{SD}(\rho) = 1/\sqrt{n-1} = 0.45$. But it need not be computed within a row: pooling over the 12 base-by-cohort blocks that already exist gives a standard error of about 0.17, and the answer is then visible. On grade-school mathematics there is no relationship, $\rho = -0.14$ with a block-bootstrap interval of -0.46 to $+0.21$. On code generation $\rho = +0.45$, interval $+0.10$ to $+0.73$, and the sign is the wrong one: *higher* NLL excess goes with a *higher* pass rate. It survives dropping the one arm with a rank confound ($\rho = +0.48$).
There is a reading of this that is not damning, and the authors give it without leaning on it: worst-task excess is a maximum over four tasks, so it penalises exactly those merges that sacrifice one task to preserve another, and a merge that protects the code adapter at the expense of translation will look bad by their metric and good on a code benchmark. That is a coherent story, and it is also an argument that the metric answers a different question from the one a practitioner asks about a single deployed capability.
The scope of this limitation is narrower than it was. What fails above is the ranking of merge *methods* within a cohort, where the differences are hundredths of a nat. The paper's own effects are not of that size, and both halves of the conditioning result are now measured in accuracy as well, which is what makes this limitation survivable.
The positive half: §5.8 finds the collapse reaching GSM8K and HumanEval on four of four bases, with the Mistral unregularised merge scoring zero on both and recovering to 0.462 and 0.244 with one ridge term. The null half: §5.2 finds no effect in any of forty accuracy cells, matching what the loss metric says about the same methods. The authors name the second explicitly because a null carried only in a disputed unit is worth much less than an effect carried in it, and reporting the flattering half twice and the null once would have been the more comfortable choice.
So a claim about method rankings in this unit should still not be read as a claim about accuracy. A claim about the conditioning regime may be, in either direction, and that is the claim this paper makes.

(3) **The finite-rate exponent is unmeasured, not merely unresolved.** Against matched asymptotes one base fits at -2.10 ($R^2 = 0.98$) and one at -3.50, but neither fit is supported at the precision this metric provides: the point that makes the first fit possible sits 0.0000029 nats above its asymptote, some 350 times below the metric's own resolution, and two of the second's three points are also below it. Ten of twenty-four measured values fall below their own asymptote, which the model forbids. The authors withdraw the reading that the exponent holds on any base (§5.9); the sweep lacks the dynamic range to measure it, since only $b \in \{1, 2\}$ clears the noise floor and $b = 1$ is excluded as clipping-dominated. A finer low-rate grid would settle it; the authors deliberately have not chosen one after seeing which would produce a fit. *No claim in this paper depends on the rate term.*

(4) **The authors' KnOTS baseline was a no-op, and the repair works.** With the default inner combination it reduces algebraically to task arithmetic and matched it to four decimals; §5.2 measures it at amplification 0.99 with tail fraction 0.000, identical to task arithmetic in both columns. The authors report that as an erratum about their own configuration and claim nothing about published KnOTS. Re-run with a TIES inner combination it becomes a genuinely different method, and a good one: it beats task arithmetic on all four bases in both cohorts, by 0.020 to 0.080 nats, and at $T = 3$ on three of four, Llama's margin falling inside the tie threshold. So a subspace-alignment method *is* represented in the authors' comparison after all. It does not reach the champion baseline on any base at either T, ranking third or fourth of six throughout, so no verdict in Table 4 moves.

(5) **One adapter in four never learned its task**, on every base and in both arms, so the authors' cohorts are $T = 4$ nominal and 3 effective (§4.7). Every geometric quantity is reported at both T and the contrast is unchanged. The merge results are now reported at both T as well, and there one verdict does move: the authors' encoder's win on Llama becomes a tie once the untrained adapter is removed (§5.1), so its remaining advantage is one of the things this defect was supporting. No claim about merging T well-trained tasks is supported by this data.

(6) **The ridge sweep's gate was one-sided in provenance, and no longer is.** With one cohort per arm the pre-registered $2 \times \mathrm{SE}$ gate cannot be computed at all, and with three cohorts on the independent side and one on the shared side its standard error comes entirely from the independent arm. Both arms now have three (§5.3), so the gate is two-sided throughout.

(7) **Train and evaluation splits overlap** by 12.9 to 16.2% on the instruction task and 9.2 to 11.6% on the code task, because the two draws shuffle one pool with different seeds. Absolute numbers are optimistic and no number here is a held-out number. Whether that affects the *comparisons* is no longer a matter of belief: re-scoring every cell on the disjoint subset leaves the method ordering unchanged in 19 of 20 rows, and the one change swaps two variants of the authors' own encoder (Reproducibility Statement).

(8) **Theorem 4's rate term is conditional.** It holds under Assumption 3, which the authors state rather than prove: that the entropy-power-to-variance ratio of η is bounded below uniformly in (T, r, d) and in the spectra. Appendix B isolates the single step that uses it. The floor term carries no such condition.

(9) **Scope.** $T = 4$, $r = 16$, four base models in the 7B to 9B range, language only, worst-task NLL excess. $Tr = 64$ is far below d, so the regime where Tr approaches d, and where $d_{\text{eff}} = Tr$ must fail, is untouched. No rank sweep and no T sweep on real adapters.

(10) **One pre-registered control has a known specification defect** (§5.7), which the authors did not act on, because acting on it after seeing which way it fell is what pre-registration exists to prevent.

Implications for practice. The recommendation is conditional, and it is about worst-task NLL excess, which limitation 2 says does not predict downstream accuracy. The authors state both halves rather than the convenient one.

If you merge with a method that solves a linear system, audit your cohort's conditioning and regularise. Unregularised, the same merge is 65 times worse on an ill-conditioned cohort at rank 16 and 117 times without truncation, on two solvers, one of which the authors did not build, and one ridge term removes the difference. This is the recommendation the authors are most confident in, because it is the one with a measured mechanism (Table 6) behind it.

If you merge with task arithmetic, TIES, DARE or a similar heuristic, the authors find no effect of cohort conditioning larger than about 0.01 nats, which is what their design can detect. The exception is subspace alignment, which gains 0.010 to 0.013 nats from independent initialisation on three of four bases, small but consistent.

For anyone benchmarking merging methods: report $\kappa(\bar{H})$ and the principal cosines alongside the results, since they cost a CPU minute and change how a solve-based result should be read; report more than one initialisation draw, because single-draw rankings are not reproducible; and do not use worst-task NLL excess as a proxy for deployed capability, on the authors' evidence, since on one of the two benchmarks they tested it ranks methods in the wrong order.

Future work. The benchmark-confounding claim would be settled by Part B of the prevalence registration, auditing the cohorts published merging benchmarks actually release; the authors' margin-aware control is not a substitute, for the reason §5.7 gives. A finer low-rate grid would settle the exponent. Checking whether the reference KnOTS implementation shares the degeneracy the authors' did would establish whether that finding is about the field or only about their code. And the regime where Tr approaches d, where the floor must stop being vacuous, is where this bound would start to have operational content; reaching it needs either many more tasks or much higher rank.

Reproducibility Statement. All merging methods, the evaluation harness, the geometry audit, the pre-registration documents and the analysis scripts are released with the paper. §F.4 summarises all twelve pre-registrations, and the public repository carries them in full and unedited. Each was committed to version control before the compute it governs was dispatched; the commit ordering is recoverable from the repository history and is the audit trail the authors ask the reader to check rather than take on trust.

Data splits. Training and evaluation splits are *not* disjoint. Training shuffles the source pool with the cohort seed while evaluation shuffles with a separate fixed seed, so on the two tasks whose evaluation split is the training split the held-out thousand is a different permutation's slice of the same pool. Measured directly, the overlap is 12.9 to 16.2% of evaluation examples on the instruction-following task and 9.2 to 11.6% on the code task, varying by cohort. Grade-school mathematics and translation draw from genuinely separate splits and are unaffected. Absolute loss and accuracy numbers in this paper are therefore optimistic and should not be read as held-out performance.

Whether that overlap affects comparisons *between* methods is testable rather than a matter of belief, and the authors tested it. Every cell stores per-example loss in evaluation order, so the contaminated positions can be dropped and the whole matrix re-aggregated without retraining anything. On the disjoint subset the method ordering is unchanged in 19 of the 20 base-by-cohort rows; the single exception swaps two variants of the authors' own encoder with each other and moves no baseline; and the worst task is the same in all 144 cells. Per-cell worst-task excess shifts by a mean of 0.0000 nats, between −0.0021 and +0.0009. The claim that the overlap does not affect the authors' comparisons is therefore a result rather than an assurance, and it is worth being precise about what it covers: comparisons, not levels. Absolute numbers remain optimistic.

Evaluation harnesses. The downstream GSM8K and HumanEval numbers in this paper are re-scored. The original scorers discarded a large and method-dependent fraction of generations: the HumanEval extractor returned an empty completion for markdown-fenced output, and the GSM8K extractor required the answer at end-of-string. Discard rates ran from 61% to 81% on some method and base combinations and near zero on others, which makes cross-method comparison meaningless. Both scorers have been fixed, all affected cells were re-scored, and the discard rate is now zero. The re-scored numbers are what appear here. The authors do not draw any correlation conclusion from them, for the power reasons given in §6.

Code and data. The merge implementations, the evaluation harness, the geometry audit, every pre-registration document, and the per-cell result files are public at https://github.com/K144U/Initialisation-Sets-the-Conditioning-of-LoRA-Merging. It is a repository rather than a snapshot: clone it and the commit graph is intact, so the ordering claimed in Table 17 can be checked with `git merge-base` rather than taken on trust. That ordering is the part of the authors' procedure they are asking the reader to verify, and the hashes in that table are this repository's.

The published history is filtered, and the authors say so because the filtering rewrote every commit hash. Three things are absent from it, none of them cited anywhere in this paper and none bearing on any result: private correspondence and external review notes, which were never the authors' alone to publish; large build artifacts, which only bloat the history; and text identifying a manuscript that is still under anonymous review elsewhere. Filtering changes hashes but not the commit graph, and the graph is what the verification depends on.

The base model weights and the trained adapters are not redistributed: the bases are public checkpoints, and the adapter training configurations, including every seed, are in the repository.

Compute. All merging and evaluation ran on a single node with A100-80GB GPUs, one evaluation cell per card at 18 to 21 GiB. The project as a whole trained 112 LoRA adapters and completed roughly 1,200 evaluation cells at 17 to 26 minutes each, on the order of 400 GPU-hours, of which the experiments reported in this paper are a subset (Appendix G). That total includes superseded runs: the results this paper reports were not obtained on the first attempt, and several of the corrections described in §6 required re-running cells that had already been analysed.

The geometry and conditioning audit of §4.5 is the exception and is deliberately cheap: it runs on CPU in about one minute per cohort from the adapter weight files alone, with no forward passes, so that measurement and every number in Tables 1 and 2 are reproducible without GPU access.

Seeds and provenance. Cohort initialisation seeds, evaluation data seeds, and merge-time mask seeds are independent and are recorded per cell in the released manifests. The distinction matters in this paper more than usual: §4 turns entirely on whether the initialisation seed is shared across the tasks of a cohort, and the shared-seed condition is the one that produces the collapsed regime.

Acknowledgments

The authors are thankful to Jaypee Edusphere for providing access to its supercomputing facility, Ramanujan Universe, on which the roughly 400 GPU-hours of A100-80GB compute reported here were carried out. The authors also thank the anonymous reviewers of earlier versions of this manuscript, whose criticism led directly to the reorganisation of the theoretical material and to the pre-registered replication reported in §5.

References

S. Ashkboos, A. Mohtashami, M. L. Croci, B. Li, P. Cameron, M. Jaggi, D. Alistarh, T. Hoefler, and J. Hensman. 2024. "QuaRot: Outlier-Free 4-Bit Inference in Rotated LLMs." In: *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv: 2404.00456.

Y. Cao, D. Ran, Y. Guo, M. Wu, S. Chen, L. Li, W. Yang, and T. Xie. 2026. "An Empirical Study and Theoretical Explanation on Task-Level Model-Merging Collapse." *arXiv preprint arXiv:2603.09463*. arXiv: 2603.09463.

T. M. Cover and J. A. Thomas. 2006. *Elements of Information Theory*. (2nd ed.). Wiley-Interscience.

P. T. Deep, R. Bhardwaj, and S. Poria. 2024. "DELLA-Merging: Reducing Interference in Model Merging through Magnitude-Based Sampling." *arXiv preprint arXiv:2406.11617*. arXiv: 2406.11617.

A. A. El Gamal and T. M. Cover. 1982. "Achievable Rates for Multiple Descriptions." *IEEE Transactions on Information Theory*, 28, 6, 851–857.

O. K. Hitit, L. Girrbach, and Z. Akata. 2025. "A Systematic Study of Model Merging Techniques in Large Language Models." *arXiv preprint arXiv:2511.21437*. arXiv: 2511.21437.

E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. 2022. "LoRA: Low-Rank Adaptation of Large Language Models." In: *International Conference on Learning Representations (ICLR)*.

X. Huang, Z. Liu, S.-Y. Liu, and K.-T. Cheng. 2024. "RoLoRA: Fine-tuning Rotated Outlier-free LLMs for Effective Weight-Activation Quantization." In: *Findings of the Association for Computational Linguistics: EMNLP 2024*, 7563–7576. arXiv: 2407.08044.

G. Ilharco, M. T. Ribeiro, M. Wortsman, L. Schmidt, H. Hajishirzi, and A. Farhadi. 2023. "Editing Models with Task Arithmetic." In: *International Conference on Learning Representations (ICLR)*.

W. Jeong, W. Lee, and K.-J. Yoon. 2026. "Preference-Aligned LoRA Merging: Preserving Subspace Coverage and Addressing Directional Anisotropy." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. arXiv: 2603.26299.

X. Jin, X. Ren, D. Preoţiuc-Pietro, and P. Cheng. 2023. "Dataless Knowledge Fusion by Merging Weights of Language Models." In: *International Conference on Learning Representations (ICLR)*. arXiv: 2212.09849.

Y. Kim, S. Lee, A. Jung, B. Ryu, and S. Hong. 2025. "Task Vector Quantization for Memory-Efficient Model Merging." In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. arXiv: 2503.06921.

Z. Li, X. Feng, W. Ma, X. Feng, and B. Qin. 2025. *Leveraging Rotation Symmetry for Efficient LoRA Merging in Large Language Models*. OpenReview submission to ICLR 2026, withdrawn; forum id `iE0dWuv6jU`. (2025).

S. Liu, Y. Yao, B. He, Z. Liu, X. Han, M. Yuan, H. Wu, and L. Song. 2025. "1bit-Merging: Dynamic Quantized Merging for Large Language Models." *arXiv preprint arXiv:2502.10743*. arXiv: 2502.10743.

Z. Liu, C. Zhao, I. Fedorov, B. Soran, D. Choudhary, R. Krishnamoorthi, V. Chandra, Y. Tian, and T. Blankevoort. 2025. "SpinQuant: LLM Quantization with Learned Rotations." In: *The Thirteenth International Conference on Learning Representations (ICLR)*. arXiv: 2405.16406.

M. Matena and C. Raffel. 2022. "Merging Models with Fisher-Weighted Averaging." *Advances in Neural Information Processing Systems (NeurIPS)*.

T.-H. Nguyen, D. Huu-Tien, T. Suzuki, and L.-M. Nguyen. 2026. "RegMean++: Enhancing Effectiveness and Generalization of Regression Mean for Model Merging." *Transactions on Machine Learning Research (TMLR)*. arXiv: 2508.03121.

G. Ortiz-Jimenez, A. Favero, and P. Frossard. 2023. "Task Arithmetic in the Tangent Space: Improved Editing of Pre-Trained Models." In: *Advances in Neural Information Processing Systems (NeurIPS) Oral*.

A. Panariello, D. Marczak, S. Magistri, A. Porrello, B. Twardowski, A. D. Bagdanov, S. Calderara, and J. van de Weijer. 2025. "Accurate and Efficient Low-Rank Model Merging in Core Space." In: *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv: 2509.17786.

S. Pathak and S. Garg. 2026. *A Rate-Distortion Function for Model Merging*. Zenodo. Version v3. Preprint. Superseded by the present paper. (2026). doi:10.5281/zenodo.21238820.

R. Salami, P. Buzzega, M. Mosconi, J. Bonato, L. Sabetta, and S. Calderara. 2024. "Closed-Form Merging of Parameter-Efficient Modules for Federated Continual Learning." *arXiv preprint arXiv:2410.17961*. Introduces LoRM; setting is federated continual learning. arXiv: 2410.17961.

C. E. Shannon. 1959. "Coding Theorems for a Discrete Source with a Fidelity Criterion." *IRE National Convention Record*, 7, 142–163.

G. Stoica, P. Ramesh, B. Ecsedi, L. Choshen, and J. Hoffman. 2025. "Model Merging with SVD to Tie the KnOTS." In: *The Thirteenth International Conference on Learning Representations (ICLR)*. arXiv: 2410.19735.

A. Tseng, J. Chee, Q. Sun, V. Kuleshov, and C. De Sa. 2024. "QuIP#: Even Better LLM Quantization with Hadamard Incoherence and Lattice Codebooks." In: *Proceedings of the 41st International Conference on Machine Learning (ICML)* (Proceedings of Machine Learning Research). Vol. 235, 48630–48656. arXiv: 2402.04396.

P. Yadav, D. Tam, L. Choshen, C. Raffel, and M. Bansal. 2023. "TIES-Merging: Resolving Interference When Merging Models." In: *Advances in Neural Information Processing Systems (NeurIPS)*.

E. Yang, Z. Wang, L. Shen, S. Liu, G. Guo, X. Wang, and D. Tao. 2024. "AdaMerging: Adaptive Model Merging for Multi-Task Learning." In: *International Conference on Learning Representations (ICLR)*. arXiv: 2310.02575.

A. C.-C. Yao. 1977. "Probabilistic Computations: Toward a Unified Measure of Complexity." In: *18th Annual Symposium on Foundations of Computer Science (FOCS)*, 222–227.

Y. Yao et al.. 2026. "Merging Beyond: Streaming LLM Updates via Activation-Guided Rotations." *arXiv preprint arXiv:2602.03237*. arXiv: 2602.03237.

L. Yu, B. Yu, H. Yu, F. Huang, and Y. Li. 2024. "Language Models are Super Mario: Absorbing Abilities from Homologous Models as a Free Lunch (DARE)." In: *International Conference on Machine Learning (ICML)*.

A. Zandieh, M. Daliri, M. Hadian, and V. Mirrokni. 2025. "TurboQuant: Online Vector Quantization with Near-optimal Distortion Rate." *arXiv preprint arXiv:2504.19874*. arXiv: 2504.19874.

S. Zeng, Y. He, M. Liu, W. You, Y. Hao, Y.-H. H. Tsai, M. Yamada, and H. Zhao. 2025. "Task Vector Bases: A Unified and Scalable Framework for Compressed Task Arithmetic." *arXiv preprint arXiv:2502.01015*. arXiv: 2502.01015.

S. Zheng et al.. 2025. "Decouple and Orthogonalize: A Data-Free Framework for LoRA Merging." *arXiv preprint arXiv:2505.15875*. arXiv: 2505.15875.

L. Zhou, D. Solombrino, D. Crisostomi, M. S. Bucarelli, G. A. D'Inverno, F. Silvestri, and E. Rodolà. 2025. "On Task Vectors and Gradients." *arXiv preprint arXiv:2508.16082*. arXiv: 2508.16082.

L. Zhou, D. Solombrino, D. Crisostomi, M. S. Bucarelli, F. Silvestri, and E. Rodolà. 2024. "ATM: Improving Model Merging by Alternating Tuning and Merging." *arXiv preprint arXiv:2411.03055*. arXiv: 2411.03055.

A The Rate-Distortion Bound

The main text's §3 states what this bound did and did not do for the paper. This appendix gives the formal development: the coding model, the hard distribution, and the two theorems. Proofs are in Appendix B.

A.1 Rate-R merging codes

A rate-R merging code is a pair (E, D) with encoder $E : (\mathbb{R}^d)^T \to \{1, \dots, 2^R\}$ and decoder $D : \{1, \dots, 2^R\} \to \mathbb{R}^d$, both of which may depend on $\boldsymbol{H}$ as shared side information: in deployment $\boldsymbol{H}$ is determined by the LoRA factors both ends already hold, so only the R bits describing $\boldsymbol{\tau}$ are charged to the rate. The authors' lower bound holds even when $\boldsymbol{H}$ is revealed for free to both ends. The *rank-r rate-distortion function* is

$$\mathcal{D}^\star(T, d, r, B, R) \;:=\; \inf_{(E,D)} \; \sup_{(\boldsymbol{\tau},\boldsymbol{H})\text{ admissible}} \mathbb{E}\big[\Delta(w^\star; \boldsymbol{\tau}, \boldsymbol{H})\big], \qquad w^\star := D(E(\boldsymbol{\tau})), \tag{5}$$

with the expectation over any shared randomness, and Δ the worst-task distortion (2). In the floor-zero Stiefel regime this appendix determines $\mathcal{D}^\star$ up to a universal constant: a lower bound for every code (Theorem 4) and an explicit code matching it (Theorem 5).

Hard distribution $P^\star$. The lower bound is established against a distribution $P^\star$ on admissible tuples. For each task t, independently: (i) sample $U_t \in \mathrm{Stiefel}(d, r)$ uniformly and set $V_t := \mathrm{range}(U_t)$; (ii) fix a positive-definite diagonal $D_t \succ 0$ (the task-t spectrum) and set $H_t := U_t D_t U_t^\top$; (iii) sample $u_t \sim \mathrm{Unif}(S^{r-1})$ and set $\tau_t := U_t\, B D_t^{-1/2} u_t$. The normalisation $\tau_t^\top H_t \tau_t = B^2$ holds by construction, and the $D_t^{-1/2}$ stretch makes the second-moment identity $\mathbb{E}[H_t \tau_t \tau_t^\top H_t \mid U_t] = (B^2/r) H_t$ hold *regardless of* D_t, the feature that makes Theorem 4 apply to arbitrary task-dependent spectra.

Yao's minimax bridge. By Yao's minimax principle (A. C.-C. Yao 1977), $\mathcal{D}^\star$ is at least the Bayesian rate-distortion function under the prior $P^\star$:

$$\mathcal{D}^\star(T, d, r, B, R) \;\geq\; \inf_{(E,D)} \; \mathbb{E}_{(\boldsymbol{\tau},\boldsymbol{H})\sim P^\star}\big[\Delta(w^\star; \boldsymbol{\tau}, \boldsymbol{H})\big] \tag{6}$$

All lower bounds below are proved against the right-hand side.

A.2 The lower bound

The lower bound rests on three ingredients: a max-$\geq$-avg identity converting max-distortion into average distortion plus an irreducible floor; a closed-form evaluation of that floor under $P^\star$; and a Shannon converse on the remaining compression subproblem, conditional on the curvature data $\boldsymbol{H}$. Lemmas 1 and 2 are proved in full in Appendix B. Theorem 4's floor term is proved there unconditionally; its rate term rests on Assumption 3 below, which the authors state explicitly rather than leave in an appendix.

Lemma 1 (H_t-weighted max $\geq$ avg identity). *For every admissible $(\boldsymbol{\tau}, \boldsymbol{H})$ and every $w \in \mathbb{R}^d$,*

$$\frac{1}{T}\sum_{t=1}^{T}(w - \tau_t)^\top H_t (w - \tau_t) \;=\; (w - \bar{\tau}_H)^\top \bar{H}(w - \bar{\tau}_H) + \frac{1}{T}\sum_{t=1}^{T}(\tau_t - \bar{\tau}_H)^\top H_t(\tau_t - \bar{\tau}_H), \tag{7}$$

and hence $\max_t (w - \tau_t)^\top H_t (w - \tau_t)$ *is at least the right-hand side.*

The proof is a two-line expansion: the cross-term vanishes by the defining property $\bar{H}\bar{\tau}_H = T^{-1}\sum_t H_t \tau_t$ of the centroid. Two consequences drive everything that follows: components of w orthogonal to $\mathrm{range}(\bar{H}) = V_1 + \cdots + V_T$ annihilate against every H_t, so without loss the decoder's output lies in an at-most-d_{eff}-dimensional subspace; and the bound separates into a compression term in the $\bar{H}$-metric on $\bar{\tau}_H$ and a floor term that does not depend on w at all.

Lemma 2 (Closed-form floor). *Under $P^\star$ with any fixed positive-definite spectra $(D_t)_{t=1}^T$ (possibly distinct across tasks),*

$$\mathbb{E}_{P^\star}\left[\frac{1}{T}\sum_{t=1}^{T}(\tau_t - \bar{\tau}_H)^\top H_t(\tau_t - \bar{\tau}_H)\right] \;=\; B^2\left(1 - \frac{\mathbb{E}[d_{\mathrm{eff}}]}{Tr}\right). \tag{8}$$

The key probabilistic input is the second-moment identity $\mathbb{E}[H_t\tau_t\tau_t^\top H_t \mid U_t] = (B^2/r)H_t$, which holds for any D_t by the $D_t^{-1/2}$ stretch in $P^\star$. Three limits clarify the formula: at $r = d$ (full rank) the floor is $B^2(1 - 1/T)$, the classical averaged-vector-quantisation spread; at $T = 1$ it is zero, recovering single-source quantisation; and when all subspaces coincide ($V_t = V$, $d_{\text{eff}} = r$) it is again $B^2(1 - 1/T)$. In the generic LoRA regime (V_t Stiefel-random, $Tr \leq d$), $d_{\text{eff}} = Tr$ almost surely and the floor vanishes.

The rate term needs one ingredient the authors do not prove, and they state it as an assumption rather than carry it as a caveat, so that what is conditional is visible where the result is used.

Assumption 3 (Uniform entropy power). *There is an absolute constant $c_{\text{TQ}} > 0$, independent of (T, r, d) and of the spectra $(D_t)_{t=1}^T$, such that the conditional entropy power of $\eta := \bar{H}^{1/2}\bar{\tau}_H$ given $\boldsymbol{H}$ is at least c_{TQ} times its conditional variance.*

For each fixed instance the ratio is finite, because η has a density and its scale is pinned by $\tau_t^\top H_t\tau_t = B^2$. What the authors do not establish is that the bound is uniform, and Appendix B isolates the exact step where it is used. The authors import the constant from TurboQuant (Zandieh et al. 2025), which supplies one in the rotation-invariant case; transferring it to their conditional setting is part of what Assumption 3 asserts.

Theorem 4 (Rank-r RD lower bound, general H_t). *Under $P^\star$ with any fixed positive-definite spectra $(D_t)_{t=1}^T$ and $\operatorname{rank}(H_t) = r$ for all t, and under Assumption 3 for the rate term,*

$$\mathcal{D}^\star(T, d, r, B, R) \;\geq\; B^2\left(1 - \frac{\mathbb{E}[d_{\text{eff}}]}{Tr}\right) \;+\; c_{\text{TQ}} \cdot \frac{B^2\mathbb{E}[d_{\text{eff}}]}{Tr} \cdot 2^{-2R/\mathbb{E}[d_{\text{eff}}]}. \tag{9}$$

Under Stiefel-random sampling with $Tr \leq d$, $d_{\text{eff}} = Tr$ a.s., the floor vanishes, and $\mathcal{D}^\star(T, d, r, B, R) \geq c_{\text{TQ}}B^2\,2^{-2R/(Tr)}$, with c_{TQ} the universal constant of TurboQuant (Zandieh et al. 2025) Theorem 3.

Proof sketch. By (6) it suffices to lower-bound the Bayesian risk under $P^\star$. Lemmas 1 and 2 reduce it to the floor plus the compression term $\mathbb{E}[(w^\star - \bar{\tau}_H)^\top\bar{H}(w^\star - \bar{\tau}_H)]$; the decoder may be assumed to emit $w^\star \in \operatorname{range}(\bar{H})$, and with $\eta := \bar{H}^{1/2}\bar{\tau}_H \in \mathbb{R}^{d_{\text{eff}}}$ the subproblem is rate-R quantisation of η under squared loss. Given $\boldsymbol{H}$, the only remaining randomness is the directions u_t, so a conditional Shannon converse applies to every rate-R code, *including codebooks adapted to $\boldsymbol{H}$*: conditioning on $\boldsymbol{H}$, rather than averaging over rotations, is what makes the converse valid against the $\boldsymbol{H}$-aware decoder of Appendix A.3; the constant absorbs the entropy-power-to-variance ratio of η, which is $O(1)$ because the scale is pinned by $\tau_t^\top H_t\tau_t = B^2$. Combining with $\mathbb{E}[\|\eta\|^2] = B^2\mathbb{E}[d_{\text{eff}}]/(Tr)$ from Lemma 2 yields (9).

On putting $\mathbb{E}[d_{\text{eff}}]$ inside the exponent. The converse is applied conditionally on $\boldsymbol{H}$, so what the argument delivers is $\mathbb{E}[g(d_{\text{eff}})]$ with $g(x) := x\,2^{-2R/x}$, whereas (9) is stated as $g(\mathbb{E}[d_{\text{eff}}])$. The substitution needs g convex, and it is, on all of $x > 0$: writing $a := 2R\ln 2$,

$$g''(x) \;=\; \frac{a^2}{x^3}\,e^{-a/x} \;>\; 0,$$

so Jensen gives $\mathbb{E}[g(d_{\text{eff}})] \geq g(\mathbb{E}[d_{\text{eff}}])$, which is the direction a lower bound requires. The step is worth spelling out because the natural misreading fails: the exponential factor *alone*, $x \mapsto 2^{-2R/x}$, has second derivative $(a/x^3)(a/x - 2)e^{-a/x}$ and is convex only for $x < R\ln 2$, so it would not license the substitution at the dimensions the authors work at. It is the linear prefactor, which comes from $\mathbb{E}[\|\eta\|^2] = B^2\mathbb{E}[d_{\text{eff}}]/(Tr)$, that makes the product convex throughout. The point is moot in the Stiefel regime, where $d_{\text{eff}} = Tr$ almost surely and the expectation is over a constant.

Why the general-H_t statement is the headline. Real LoRA fine-tunes have task-specific curvature: the Fisher information at τ_t is rank-r PSD with range V_t but generally not a projector. Theorem 4 applies with no modeling assumption on the D_t beyond positive-definiteness, closing the MSE $\rightarrow$ CE bridge for the lower-bound half.

A.3 Achievability

The lower bound is matched by an explicit algorithm in the generic regime (V_t Stiefel-random, $Tr \le d$, so $d_{\text{eff}} = Tr$ a.s. and the floor vanishes). The algorithm extends the single-source construction of TurboQuant (Zandieh et al. 2025) with a centroid-reduction step via Lemma 1.

Algorithm. Encoder and decoder share a pseudorandom seed indexing an orthogonal rotation $O \in O(d_{\text{eff}})$ drawn from Haar measure (implemented by QR of an iid Gaussian matrix). *Encoder:* (1) compute $\bar{H}$, $\bar{\tau}_H$, and an orthonormal basis $Q \in \mathbb{R}^{d \times d_{\text{eff}}}$ of range$(\bar{H})$ with eigenvalues $w_1, \ldots, w_{d_{\text{eff}}} > 0$; form $\eta := \bar{H}^{1/2}\bar{\tau}_H$ in the Q-basis. (2) Apply the shared rotation $\tilde{\eta} := O\eta_Q$. (3) Uniform-scalar-quantise each coordinate on $[-c\sigma_{\text{pc}}, c\sigma_{\text{pc}}]$ with b bits, where $\sigma_{\text{pc}} := B/\sqrt{Tr}$ and c is a clipping constant (the authors use $c = 5$); total rate $R = bd_{\text{eff}}$. *Decoder:* invert each step and emit $w^\star := \bar{H}^{+1/2}\eta_q$ on range$(\bar{H})$. The centroid $\bar{\tau}_H$ is the sufficient statistic in the floor-zero regime, the $\bar{H}^{1/2}$ scaling whitens it, and the random rotation spreads quantisation error equally across directions.

Theorem 5 (Matching achievability, floor-zero Stiefel). *Under $P^\star$ with Stiefel-random V_t and $Tr \le d$, for any fixed positive-definite $(D_t)_{t=1}^T$, the algorithm above at rate R achieves*

$$\mathbb{E}_{P^\star}\left[\max_t (w^\star - \tau_t)^\top H_t (w^\star - \tau_t)\right] \le C \cdot B^2 \cdot 2^{-2R/(Tr)}, \qquad C = Tc^2/3. \tag{10}$$

Combined with Theorem 4, the Bayesian rate-distortion function under $P^\star$ is $\Theta\big(B^2\, 2^{-2R/(Tr)}\big)$ in this regime, universal in d, r, R, B, with the T-dependence of the constant discussed in Remark 6.

Proof sketch. In the floor-zero regime Lemma 2 forces $H_t\bar{\tau}_H = H_t\tau_t$ almost surely, so worst-task distortion is driven entirely by the quantisation error $\delta\eta$ on the whitened centroid; the crude bound $\max_t \le \sum_t = T \cdot \text{avg}$ gives $\max_t(\delta w)^\top H_t(\delta w) \le T\|\delta\eta\|^2$, and uniform scalar quantisation after the rotation contributes per-coordinate MSE at most $c^2\sigma_{\text{pc}}^2 2^{-2b}/3$.

Remark 6 (What the Θ hides; the T-dependence is open). The lower-bound constant is T-free while the upper-bound constant is $C = Tc^2/3$; the entire factor T enters at the crude $\max_t \le T$ avg step, which is loose except when one task's error projection dominates, a case atypical in the floor-zero regime. The synthetic measurements decide this in favour of the benign reading: sweeping $T \in \{2, 4, 8, 16\}$ (1000 trials/cell, Appendix C), the achievability constant grows *logarithmically* ($R^2 \ge 0.998$ for a $\log T$ fit vs. ≈ 0.91 linear; the T=16 constant is 1.77× the T=2 value, against 8× if linear). The linear T is an artifact of the analysis, not intrinsic; proving the matching $\log T$ (or T-free) constant is open, and the floor-zero Θ should be read as "Θ up to a $\log T$ factor."

The shared-V / floor-positive regime. The regime below is a property of the model, not of any cohort the authors measured: §4.3 finds $d_{\text{eff}} = Tr$ on every real cohort, in both initialisation conditions, so none of them occupies it. It is reachable when the V_t genuinely coincide, which for LoRA would need $Tr > d$ or exactly shared subspaces rather than merely near-collinear ones. When they do overlap ($d_{\text{eff}} < Tr$, floor positive), the zero-floor step fails. A refinement quantising around the Chebyshev center of the ensemble in an active-set-aware null-space split achieves excess distortion of order $2^{-2R/(d_{\text{eff}}+|A|-1)}$, where $|A|$ is the active-set size: better than naive scalar quantisation but short of the lower bound's $2^{-2R/d_{\text{eff}}}$.

Remark 7 (Shared-V exponent gap). Valid rate-R random-codebook encoders empirically achieve slopes strictly steeper than the linear construction (slope −1.45 vs. the linear prediction −1.20 at $T = 3$, $r = 3$; Appendix C), so the true rate-distortion function in the floor-positive regime lies strictly between the two bounds. Neither bound is tight; the exact exponent is open.

B Proofs

This appendix supplies the proofs that §A defers. An earlier pointer sent the reader to supplementary material; no such material existed, and the authors correct that here.

The authors are explicit about what is fully proved and what is not. Lemma 1 and Lemma 2 are proved completely below. Theorem 4 is proved modulo one constant, and the authors isolate exactly which step carries it rather than absorbing it silently.

B.1 Proof of Lemma 1

Fix any $w \in \mathbb{R}^d$ and recall $\bar{H} = \frac{1}{T}\sum_t H_t$ and that $\bar{\tau}_H$ is any solution of $\bar{H}\bar{\tau}_H = \frac{1}{T}\sum_t H_t\tau_t$ lying in $\mathrm{range}(\bar{H})$. Decompose $w - \tau_t = (w - \bar{\tau}_H) + (\bar{\tau}_H - \tau_t)$ and expand the quadratic form:

$$(w-\tau_t)^\top H_t(w-\tau_t) = (w-\bar{\tau}_H)^\top H_t(w-\bar{\tau}_H) + 2(w-\bar{\tau}_H)^\top H_t(\bar{\tau}_H-\tau_t) + (\bar{\tau}_H-\tau_t)^\top H_t(\bar{\tau}_H-\tau_t).$$

Average over $t \in [T]$. The first term averages to $(w-\bar{\tau}_H)^\top \bar{H}(w-\bar{\tau}_H)$ by linearity. The cross term averages to

$$2(w-\bar{\tau}_H)^\top\left[\bar{H}\bar{\tau}_H - \frac{1}{T}\sum_t H_t\tau_t\right] = 0,$$

which vanishes by the defining property of $\bar{\tau}_H$ and is the only place that property is used. The third term does not involve w. This gives (7). Since a maximum is at least an average, $\max_t (w-\tau_t)^\top H_t(w-\tau_t)$ is at least the right-hand side. □

Two consequences used later. First, any component of w orthogonal to $\mathrm{range}(\bar{H}) = V_1 + \cdots + V_T$ annihilates against every H_t, so a decoder may be assumed without loss to emit $w \in \mathrm{range}(\bar{H})$, a space of dimension d_{eff}. Second, the bound separates: a w-dependent compression term in the $\bar{H}$ metric, and a w-independent floor.

An exact instance form. Expanding the floor and using $\frac{1}{T}\sum_t H_t\tau_t = \bar{H}\bar{\tau}_H$ and $\tau_t^\top H_t\tau_t = B^2$ under $P^\star$,

$$\frac{1}{T}\sum_t(\tau_t-\bar{\tau}_H)^\top H_t(\tau_t-\bar{\tau}_H) \;=\; B^2 - \bar{\tau}_H^\top\bar{H}\bar{\tau}_H. \tag{11}$$

This form is worth stating because it is computable for any given cohort from the adapters alone, with no appeal to $P^\star$. The authors use it in §4, and they note that it is the quantity a practitioner should evaluate; the prior-averaged expression of Lemma 2 is a statement about $P^\star$, not about the cohort in front of them.

B.2 Proof of Lemma 2

The authors first record the second-moment identity that makes the result independent of the spectra D_t.

Claim 8. *Under $P^\star$, for every t, $\mathbb{E}[H_t\tau_t\tau_t^\top H_t \mid U_t] = (B^2/r)\,H_t$, and $\tau_t^\top H_t\tau_t = B^2$ almost surely.*

Proof. With $H_t = U_tD_tU_t^\top$ and $\tau_t = U_tBD_t^{-1/2}u_t$, using $U_t^\top U_t = I_r$,

$$H_t\tau_t = U_tD_tU_t^\top U_tBD_t^{-1/2}u_t = B\,U_tD_t^{1/2}u_t.$$

Hence $H_t\tau_t\tau_t^\top H_t = B^2U_tD_t^{1/2}u_tu_t^\top D_t^{1/2}U_t^\top$. Since $u_t \sim \mathrm{Unif}(S^{r-1})$ has $\mathbb{E}[u_tu_t^\top] = I_r/r$, taking the conditional expectation gives $(B^2/r)\,U_tD_tU_t^\top = (B^2/r)H_t$. For the second statement, $\tau_t^\top H_t\tau_t = B^2u_t^\top D_t^{-1/2}D_tD_t^{-1/2}u_t = B^2\|u_t\|^2 = B^2$. The $D_t^{-1/2}$ stretch in $P^\star$ is exactly what makes both statements free of D_t. □

Now write $g := \frac{1}{T}\sum_t H_t\tau_t = \bar{H}\bar{\tau}_H$, so that by (11) the floor equals $B^2 - \bar{\tau}_H^\top\bar{H}\bar{\tau}_H$, and since $\bar{\tau}_H = \bar{H}^+g$ with $g \in \mathrm{range}(\bar{H})$,

$$\bar{\tau}_H^\top\bar{H}\bar{\tau}_H = g^\top\bar{H}^+\bar{H}\bar{H}^+g = g^\top\bar{H}^+g.$$

Condition on $\boldsymbol{H}$, equivalently on $(U_t, D_t)_t$. The u_t are independent across t with $\mathbb{E}[u_t] = 0$, so $\mathbb{E}[H_t \tau_t] = 0$ and the cross terms vanish:

$$\mathbb{E}[gg^\top \mid \boldsymbol{H}] = \frac{1}{T^2} \sum_{t=1}^{T} \mathbb{E}[H_t \tau_t \tau_t^\top H_t \mid \boldsymbol{H}] = \frac{1}{T^2} \sum_{t=1}^{T} \frac{B^2}{r} H_t = \frac{B^2}{rT} \bar{H},$$

using Claim 8. Therefore

$$\mathbb{E}\big[g^\top \bar{H}^+ g \mid \boldsymbol{H}\big] = \operatorname{tr}\big(\bar{H}^+ \mathbb{E}[gg^\top \mid \boldsymbol{H}]\big) = \frac{B^2}{rT} \operatorname{tr}(\bar{H}^+ \bar{H}) = \frac{B^2}{rT} \operatorname{rank}(\bar{H}) = \frac{B^2 d_{\text{eff}}}{Tr},$$

where $\operatorname{tr}(\bar{H}^+ \bar{H}) = \operatorname{rank}(\bar{H})$ because $\bar{H}^+ \bar{H}$ is the orthogonal projector onto $\operatorname{range}(\bar{H})$, and $\operatorname{rank}(\bar{H}) = \dim(V_1 + \cdots + V_T) = d_{\text{eff}}$ since the H_t are PSD, so no cancellation can occur in the sum. Taking expectations over $\boldsymbol{H}$ and substituting into (11),

$$\mathbb{E}_{P^\star}\Big[\tfrac{1}{T} \sum_t (\tau_t - \bar{\tau}_H)^\top H_t (\tau_t - \bar{\tau}_H)\Big] = B^2 - \frac{B^2\, \mathbb{E}[d_{\text{eff}}]}{Tr} = B^2\Big(1 - \frac{\mathbb{E}[d_{\text{eff}}]}{Tr}\Big). \qquad \square$$

Remark: the floor is vacuous for LoRA cohorts in general position. The floor vanishes precisely when $d_{\text{eff}} = Tr$, that is when the V_t are in direct sum. Under Stiefel-random sampling with $Tr \le d$ this holds almost surely, which is the regime the theorem's second display assumes. It also holds for every real cohort the authors measured, including the shared-initialisation ones whose subspaces are numerically near-collinear (§4). Near-collinear is not collinear: the V_t remain independent, $d_{\text{eff}} = Tr$ exactly, and by (11) the exact floor is zero. What degrades in that regime is not the floor but the conditioning of $\bar{H}$, and hence the norm of the interpolator achieving it. The authors regard this as the main practical lesson of the lower bound and they develop it in §4.

B.3 Proof of Theorem 4, and what it rests on

By Yao's principle (6) it suffices to lower-bound the Bayes risk under $P^\star$. By Lemma 1, for any code with output $w^\star$,

$$\mathbb{E}\big[\max_t D_t(w^\star)\big] \;\ge\; \underbrace{B^2\Big(1 - \frac{\mathbb{E}[d_{\text{eff}}]}{Tr}\Big)}_{\text{Lemma 2}} + \mathbb{E}\big[(w^\star - \bar{\tau}_H)^\top \bar{H} (w^\star - \bar{\tau}_H)\big].$$

It remains to lower-bound the second term. As noted, the decoder may be assumed to emit $w^\star \in \operatorname{range}(\bar{H})$. Put $\eta := \bar{H}^{1/2} \bar{\tau}_H$ and $\hat{\eta} := \bar{H}^{1/2} w^\star$, both in a space of dimension d_{eff}; the term equals $\mathbb{E}\|\hat{\eta} - \eta\|^2$, so the subproblem is rate-R quantisation of η under squared error.

Condition on $\boldsymbol{H}$. Given $\boldsymbol{H}$ the only remaining randomness is the directions u_t, and a conditional Shannon converse therefore applies to every rate-R code, *including codebooks chosen with knowledge of* $\boldsymbol{H}$. This is the step that makes the bound valid against the $\boldsymbol{H}$-aware decoder of Appendix A.3: the authors do not average over rotations, they condition. The standard converse for a d_{eff}-dimensional source under squared error gives

$$\mathbb{E}\big[\|\hat{\eta} - \eta\|^2 \,\big|\, \boldsymbol{H}\big] \;\ge\; d_{\text{eff}} \cdot \frac{2^{2h(\eta \mid \boldsymbol{H})/d_{\text{eff}}}}{2\pi e} \cdot 2^{-2R/d_{\text{eff}}},$$

with h the differential entropy. From Lemma 2, $\mathbb{E}[\|\eta\|^2 \mid \boldsymbol{H}] = B^2 d_{\text{eff}}/(Tr)$, so the per-coordinate scale is pinned.

The constant, isolated. This is the step, and the only step, that uses Assumption 3. Writing the entropy power of η as a constant fraction of its variance turns the display above into

$$c_{\text{TQ}} \cdot \frac{B^2\, \mathbb{E}[d_{\text{eff}}]}{Tr} \cdot 2^{-2R/\mathbb{E}[d_{\text{eff}}]},$$

which is (9). The step the authors do not prove here is that the entropy-power-to-variance ratio of η is bounded below by an absolute constant, uniformly in (T, r, d) and in the spectra D_t. It is finite for each fixed instance because η has a density, and the scale is pinned by $\tau_t^\top H_t \tau_t = B^2$; what the authors do not establish is uniformity. The authors therefore quote c_{TQ} from TurboQuant (Zandieh et al. 2025), Theorem 3, which supplies exactly such a constant for the rotation-invariant case, and they flag that transferring it to their conditional setting is an assumption rather than a consequence of the argument above.

Scope of the theorem. With that caveat, Theorem 4 should be read as: the floor term is exact and unconditional (it is Lemma 2), and the rate term has the stated *form* $2^{-2R/d_{\mathrm{eff}}}$ with a constant the authors import rather than derive. The authors' empirical work bears on the rate term only through §5.9, where the exponent is not resolved at the rates they could sweep. Nothing in §4 depends on the constant.

C Synthetic Validation Details

All runs use the hard distribution $P^\star$ of §3 with task-dependent spectra $(D_t)_{t=1}^T$. Code, random seeds, and raw outputs are released with the paper.

Lower-bound slope. The authors sample ensembles across $T \in \{2, 3, 4, 8\}$, $r \in \{4, 8, 16\}$, $d \in \{128, 512, 2048\}$, with both isotropic spectra ($D_t = I_r$) and four heavy-anisotropy families (geometric, linear, two-scale, and mixed; spectra spread across $[2^{-1}, 2^1]$). For each configuration the authors measure the excess $\mathbb{E}_{P^\star}[\|\bar{H}^{1/2} w^\star - \eta\|^2]$ of the Gaussian-QR + uniform scalar quantisation algorithm of Appendix A.3 as a function of $R = b d_{\mathrm{eff}}$ bits, and fit a linear regression of $\log_2 \mathbb{E}[\text{excess}]$ against $b = R/d_{\mathrm{eff}}$. Across all tested configurations the empirical slope is -2.00 ± 0.01 (95% bootstrap CI across 1000 trials per configuration), matching the prediction $-2R/d_{\mathrm{eff}}$ of Theorem 4. The floor term $B^2(1 - \mathbb{E}[d_{\mathrm{eff}}]/(Tr))$ is reproduced to four to five digits of precision in the common-D_0 and task-dependent-D_t regimes.

Achievability constant and its T-dependence. The authors measure the ratio of empirical upper bound to the Stiefel lower bound at $c_{\mathrm{TQ}} = 1$,

$$\frac{\mathbb{E}_{P^\star}\left[\max_t D_t(w^\star)\right]}{c_{\mathrm{TQ}} B^2\, 2^{-2R/(Tr)}}.$$

The ratio is constant in R (a necessary condition for the upper bound to match the lower bound in order) and starts near 11 at $T = 2$ ($r = 4$, $d = 128$), consistent with $C = Tc^2/3 \approx 17$ at $c = 5$ combined with $c_{\mathrm{TQ}} \approx 1$ from TurboQuant (Zandieh et al. 2025). Sweeping $T \in \{2, 4, 8, 16\}$ (1000 trials/cell) settles the T-dependence flagged in Remark 6: the constant grows *logarithmically*, not linearly; a $\log T$ fit gives $R^2 \geq 0.998$ against $R^2 \approx 0.91$ for a linear fit, and the T=16 value is only 1.77× the T=2 value, far below the 8× a strictly linear-T factor would give. The linear T in $C = Tc^2/3$ is thus an artifact of the crude $\max_t \leq T$ avg step, not intrinsic to the rate-distortion function; the matching $\log T$ achievability constant is stated as a conjecture.

Shared-V regime. For the floor-positive regime the authors implement the null-space-aware bit allocation of §A.3 with per-coordinate fractional bits and a locked clip $c = 11.5\sigma_{\mathrm{pc}}$ (tuned once, applied across all configurations). At $T = 2$, $r = 4$, across five anisotropy mixes (iso+iso, iso+geom, geom+twoscale, lin+twoscale, geom+iso), the empirical slope of excess over cheb^2 is -1.60 ± 0.10 at $n_{\mathrm{trials}} = 1000$, matching the theoretical prediction $-2r/(r + |A| - 1) = -8/5$ with bootstrap 95% CI ±0.010 per cell. The ±0.10 spread reflects structural anisotropy: iso+iso tracks closer to the scalar-quantisation limit -2; geom+twoscale undershoots due to H_t-metric distortion on the perpendicular subspace. Extending to $T = 3, 4$ via a SOCP Chebyshev solver with Gauss–Newton KKT refinement (residual $\sim 10^{-15}$), the null-space split continues to track the theoretical prediction across mixed anisotropies.

Table 11. Subspace geometry for all three independently initialised cohorts. The last column is $1 - d_{\text{eff}}^{\text{soft}}/(Tr)$ with $Tr = 64$. It is **not** the floor of Theorem 4, which is zero for every row here and for every row of Table 1: the theorem's d_{eff} is a rank, and substituting the soft surrogate is the error §4.3 corrects. The column is reported because it is the diagnostic that varies continuously, and to make the size of that earlier error visible. The corresponding shared-initialisation values are median cosine $\geq$ 0.9947, soft $d_{\text{eff}} \approx$ 16.3, and 0.159 to 0.191 in the fourth column. **That column is the distance between two adapters of the same cohort, not the distance either has travelled from its initialisation.** An earlier version labelled it $\|\Delta A\|/\|A\|$, which reads as the latter and contradicted the text; see §D.3 for what is and is not measured here.

base	cohort	med. cosine	$\|A_i - A_j\|_F / \|A_i\|_F$	$\sigma_{\max}$	soft d_{eff}	$1 - d_{\text{eff}}^{\text{soft}}/(Tr)$
Llama-3.1-8B	`indep1`	0.0473	1.415	1.107	63.23	0.0120
	`indep2`	0.0472	1.414	1.107	63.25	0.0118
	`indep3`	0.0467	1.415	1.105	63.24	0.0118
Mistral-7B	`indep1`	0.0468	1.412	1.106	63.24	0.0118
	`indep2`	0.0472	1.411	1.105	63.26	0.0116
	`indep3`	0.0465	1.412	1.104	63.25	0.0117
Qwen2.5-7B	`indep1`	0.0500	1.412	1.131	63.08	0.0144
	`indep2`	0.0507	1.410	1.128	63.10	0.0141
	`indep3`	0.0510	1.412	1.129	63.08	0.0143
Yi-1.5-9B	`indep1`	0.0467	1.412	1.102	63.25	0.0117
	`indep2`	0.0467	1.411	1.105	63.25	0.0117
	`indep3`	0.0473	1.412	1.105	63.25	0.0117

Random-codebook comparison. As a sanity check on the tightness of the linear construction in the shared-V regime, the authors ran a valid rate-R random-codebook encoder (2^R iid Gaussian codewords, encoder picks the one minimizing $\max_t D_t$, $R \leq 18$). At $T = 3$, $r = 3$, the random-codebook slope is -1.45, strictly steeper than the explicit linear construction's -1.20. This rules out sharpening the lower bound to match the linear upper-bound exponent; the true rate-distortion function in the floor-positive regime lies strictly between $-2r/r = -2$ and $-2r/(r + |A| - 1)$ and is left open (Remark 7).

D Subspace Geometry for All Three Independent Cohorts

Table 1 in §4.2 shows only `indep1` for each base model, on the grounds that the three independently initialised cohorts agree closely. This appendix gives all three so that claim can be checked rather than taken on trust, then repeats every geometric and conditioning quantity with the degenerate translation adapter removed, and closes with the per-adapter task performance that identifies it as degenerate.

The cohorts differ only in the random draw used to initialise the LoRA A factor of each task. The training data, data ordering seed, task configurations, rank, and number of layers sampled are identical across them. Any variation below is therefore attributable to the initialisation draw alone.

D.1 The three independent cohorts

The largest spread across cohorts on any column is 0.0010 in median cosine (Qwen), 0.005 in $\sigma_{\max}$ (Llama), and 0.02 in soft d_{eff} (Llama and Mistral). Against a shared-versus-independent difference of 0.95 in median cosine and 47 in soft d_{eff}, the between-cohort variation is smaller by three orders of magnitude.

Hard effective dimension. All twelve cells in Table 11 return $d_{\text{eff}} = Tr = 64$ at every rank tolerance in $\{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 3 \times 10^{-2}, 10^{-1}\}$, so the corresponding hard floor is exactly zero throughout. This

Table 12. Geometry at $T = 4$ (as run) and $T = 3$ (translation excluded). $\sigma_{\max}$ is compared against its collinear maximum $\sqrt{T}$, which is 2.000 at $T = 4$ and 1.732 at $T = 3$. Hard d_{eff} is at $\epsilon = 10^{-1}$ and is reported as a fraction of Tr, which is 64 at $T = 4$ and 48 at $T = 3$.

base	init	med. cosine T=4	med. cosine T=3	soft $d_{\text{eff}}/(Tr)$ T=4	soft $d_{\text{eff}}/(Tr)$ T=3	hard $d_{\text{eff}}/(Tr)$ T=4	hard $d_{\text{eff}}/(Tr)$ T=3
Llama-3.1-8B	shared	0.9948	0.9950	0.255	0.340	0.324	0.424
	independent	0.0473	0.0475	0.988	0.992	1.000	1.000
Mistral-7B	shared	0.9947	0.9945	0.254	0.339	0.311	0.409
	independent	0.0468	0.0469	0.988	0.992	1.000	1.000
Qwen2.5-7B	shared	0.9964	0.9963	0.256	0.340	0.334	0.427
	independent	0.0500	0.0502	0.986	0.990	1.000	1.000
Yi-1.5-9B	shared	0.9973	0.9970	0.254	0.338	0.316	0.406
	independent	0.0467	0.0467	0.988	0.992	1.000	1.000

coincidence is asserted programmatically in the figure-generating script, which fails rather than plots if any cell departs from 64. The shared-initialisation cohort, by contrast, returns 64 only for $\epsilon \leq 10^{-3}$ and falls to 60.5, 38.4 and 20.8 at 10^{-2}, 3×10^{-2} and 10^{-1} respectively.

D.2 Every quantity with the degenerate adapter removed

§4.7 reports that the translation adapter did not learn its task, which makes the cohorts $T = 4$ nominal and 3 effective. The tables below recompute the geometry and the conditioning with that adapter dropped, so that a reader can check directly that nothing in §4 rests on it. Nothing does: the shared-versus-independent contrast is unchanged to three decimal places in median cosine, and the independent arm continues to hold $d_{\text{eff}}^{\text{hard}} = Tr$ exactly at the new $Tr = 48$.

Soft d_{eff} is the quantity that moves, and it moves in the direction that makes the point of §4.1 rather than undermining it. In absolute terms the shared arm holds soft d_{eff} at 16.2 to 16.4 whether T is 4 or 3: the subspaces collapse onto roughly one r-dimensional subspace no matter how many tasks are stacked, so the ratio rises simply because Tr fell. The independent arm keeps soft d_{eff} at 63.1 to 63.3 of 64 and 47.5 to 47.6 of 48, which is the same fraction at both T.

Dropping the degenerate adapter lowers $\kappa(\bar{H})$ on the shared arm by 32 to 46%, which still leaves it between 1.2×10^4 and 4.8×10^4, four orders of magnitude above the independent arm's 1.4 to 1.5. The amplification tracks T on the independent arm at both values, 3.96 to 4.00 at $T = 4$ and 2.96 to 3.00 at $T = 3$, which is the orthogonal-subspace identity to within 1.4% in the worst of the eight cells here and 0.2% in the median one. Table 2 quotes 1.1% because it shows only the four $T = 4$ cells.

D.3 What the authors measure about A, and what they do not

Two different quantities can be written $\|\Delta A\|/\|A\|$ and only one of them is measured in this paper. The distinction matters because the mechanism of §4.1 is stated in terms of the one the authors do *not* measure.

Measured: the distance between two adapters of a cohort, $\|A_i - A_j\|_F/\|A_i\|_F$, the fourth column of Table 11. It is 1.410 to 1.415 on every independent cohort and 0.159 to 0.191 on the shared one. Both ends have a reference value, which makes this a useful check on the instrument. For two independent draws of equal norm, $\|A_i - A_j\|^2 \approx \|A_i\|^2 + \|A_j\|^2$, so the ratio must be about $\sqrt{2} = 1.414$; it is. At the other end, adapters that share a

Table 13. Conditioning at $T = 4$ and $T = 3$. Amplification is $\|\bar{\tau}_H\|/\|\bar{\Delta}\|$, whose value for exactly orthogonal subspaces is T.

base	init	$\kappa(\bar{H})$ T=4	$\kappa(\bar{H})$ T=3	amplification T=4	amplification T=3
Llama-3.1-8B	shared	1.76×10^4	1.20×10^4	48.3	33.7
	independent	1.54	1.42	3.99	2.99
Mistral-7B	shared	3.21×10^4	2.09×10^4	52.3	37.6
	independent	1.54	1.42	4.00	3.00
Qwen2.5-7B	shared	2.42×10^4	1.32×10^4	43.4	29.2
	independent	1.65	1.49	3.96	2.96
Yi-1.5-9B	shared	8.30×10^4	4.77×10^4	64.1	43.7
	independent	1.53	1.41	3.99	2.99

draw can differ only by what training moved, so the ratio measures that difference directly, and at 0.16 it is an order of magnitude below the independent value. §4.6 finds the same two populations in public adapters.

Also measured, on a cohort trained for the purpose: the distance each adapter travelled from its own initialisation, $\|A - A_0\|_F / \|A_0\|_F$. The cohorts the rest of this paper uses cannot answer this, because they retained neither the initial weights nor any intermediate checkpoint. Registration E3 asked for it across training and the authors reported it as unevaluable. It is evaluable on a cohort trained with the checkpoints, which is what §D.4 reports.

The two are easy to conflate and an earlier version of this work did, quoting the measured 0.16 to 0.19 as the distance each adapter had moved from its initialisation. The number is right and is the shared-cohort entry above; the description was not. The distinction decides what the number can support: a small distance *from initialisation* would say training barely moved A, which is the mechanism of §4.1, whereas a small distance *between adapters* says only that they ended near each other, which is the phenomenon. The authors have the second.

D.4 How far A actually travels

Registration E3 required $\|A - A_0\|_F / \|A_0\|_F$ at 25, 50, 75 and 100% of training, and scoped every geometry claim to "at this training budget" if it could not be evaluated. The authors report it here on a cohort trained for the purpose: one base model, both initialisation arms, four tasks each, with A_0 written before the first optimiser step and checkpoints at exactly those four fractions (steps 118, 236, 354 and 470).

Table 14. How far A moves from its initialisation, and what that does to the geometry. Median over the four tasks and over eight sampled layers, on Llama-3.1-8B. The two arms differ only in whether the four tasks drew one initialisation or four.

cohort	$\|A - A_0\|_F / \|A_0\|_F$ at 25%	50%	75%	100%	median principal cosine at 100%
shared	0.0922	0.1106	0.1262	0.1308	0.9960
independent	0.0946	0.1120	0.1257	0.1303	0.0475

A moves about 13% of its own norm over the whole of training, rising monotonically and flattening after the three-quarter mark. All three registered predictions hold, so by the registered rule **E3 is discharged** and

Table 15. Per-adapter task performance, independent arm, mean over indep1/2/3. NLL in nats on the task named in the row. The last column is the adapter that actually achieves the lowest loss on that task. The shared arm gives the identical pattern in the last column on every cell, and agrees on gain to within 1.5 percentage points; part of that difference is that two evaluation draws exist for one shared cohort, which is why the two arms are not pooled.

base	task	base NLL	specialist NLL	gain	best adapter on this task
Llama-3.1-8B	gsm8k	1.1474	0.4666	59.3%	gsm8k
	alpaca	1.2303	0.9929	19.3%	alpaca
	magicoder	0.4535	0.2814	38.0%	magicoder
	translation	0.7349	0.7468	−1.6%	gsm8k, by 0.115
Mistral-7B	gsm8k	1.3197	0.4495	65.9%	gsm8k
	alpaca	1.1323	0.8871	21.7%	alpaca
	magicoder	0.4836	0.2869	40.7%	magicoder
	translation	0.8939	0.9181	−2.7%	magicoder, by 0.210
Qwen2.5-7B	gsm8k	1.4440	0.4410	69.5%	gsm8k
	alpaca	1.2782	0.9197	28.0%	alpaca
	magicoder	0.4407	0.2477	43.8%	magicoder
	translation	1.2138	0.8219	32.3%	alpaca, by 0.019
Yi-1.5-9B	gsm8k	0.9609	0.3875	59.7%	gsm8k
	alpaca	0.9353	0.8432	9.8%	alpaca
	magicoder	0.2737	0.2295	16.1%	magicoder
	translation	0.9236	0.7589	17.8%	gsm8k, by 0.030

geometry claims are no longer scoped to an unmeasured training budget. The scope of the discharge is what the registration fixed and no more: one base model at one budget.

The control is the part worth reading. The two arms drift by 0.1308 and 0.1303, a difference of 0.0005, while their median principal cosines are 0.9960 and 0.0475, a difference of 0.948. Training moves A by the same small amount whether the cohort started from one draw or four. So the collapse is not something training does; it is the starting point, surviving a displacement that is identical in both conditions. That is the mechanism of §4.1 measured rather than argued, and it is a sharper statement than the drift number alone.

Two details the authors report rather than smooth. The final checkpoint and the saved adapter are the same weights, so the 100% column appears twice in the raw trace and the authors print it once. And the translation task drifts furthest in both arms, 0.178 and 0.183 against roughly 0.13 for the others, which is consistent with it being the adapter that never learned its task (§4.7) but which the authors did not predict and do not build on.

What the argument rests on instead is the consequence rather than the intermediate. That B starts at zero and therefore suppresses A's early gradient is a property of the parameterisation, not a measurement. That the row spaces of a shared-initialisation cohort are nearly identical is measured directly, at median principal cosine 0.995 against 0.047, and it is that measurement every downstream claim uses. The drift figure would have been a useful corroboration of why; it does not carry the claim about what.

D.5 Did the adapters learn their tasks?

Undertraining would produce a collapsed cohort too, so the two are worth distinguishing, which needs no new compute: every evaluation cell already stores the loss of the base model and of each task adapter on every task.

Table 16. Rate sweep against matched asymptotes: same ridge λ, same realisation, same cohort for the curve and its $b = \infty$ reference, with the equality asserted from the configuration files before any fit is attempted. Worst-task NLL excess in nats, at four decimals because at three the reader cannot check the two facts that matter. **Ten of the twenty-four measured values fall below their own asymptote**, which the model forbids; all but one are smaller than 10^{-4} nats and are consistent with numerical noise, the exception being Llama at $b = 3$ at -0.0020. The fit is ordinary least squares of $\log_2(\text{excess} - \text{asymptote})$ on b over $b \in \{2, 3, 4, 8\}$, keeping residuals above 10^{-9}; the "fit" column names the points that survived that filter, and both surviving fits rest partly on residuals below this metric's own resolution of about 0.001 nats (§6).

base	b=1	b=2	b=3	b=4	b=8	b=16	b=∞	fit	slope	R^2
Llama-3.1-8B	0.6329	0.0967	0.0803	0.0845	0.0823	0.0823	0.0823	2,4,8	−2.10	0.9843
Mistral-7B	0.2106	0.0497	0.0480	0.0471	0.0474	0.0473	0.0474	2,3	—	—
Qwen2.5-7B	0.0746	0.0110	0.0102	0.0101	0.0101	0.0100	0.0102	2,3	—	—
Yi-1.5-9B	0.2099	0.0413	0.0361	0.0357	0.0356	0.0355	0.0356	2,3,4	−3.50	0.9996

Three of the four tasks trained cleanly. Across both arms and all four bases, `gsm8k`, `alpaca` and `magicoder` improve on the base model by 9.8 to 69.5%, and in every one of those twenty-four cells the specialist is the best adapter on its own task. So the cohorts are not undertrained in general, and the near-collinearity of §4.2 is not an artifact of adapters that never moved.

Translation is the exception, on every base and in both arms. On Llama and Mistral the specialist is worse than the base model at its own task, by 1.6 and 2.7%. On Qwen and Yi it does improve on the base model, by 32.3 and 17.8%, but another task's adapter still beats it there, by 0.019 and 0.030 nats against a tie threshold of 0.005. By either test the adapter failed to learn a skill that distinguishes it from its cohort-mates, which is what §4.7 scopes the paper around.

E The Rate-Exponent Sweep

§5.9 reports that the authors' rate sweep cannot measure the exponent Theorem 4 predicts, and withdraws two earlier readings of it. This appendix gives the sweep, the two faults, and the registered rule applied to the result, so that a reader can check the withdrawal rather than take it.

The registered rule. Theorem 4 predicts distortion decaying as $2^{-2R/d_{\text{eff}}}$, a slope in a narrow band on a log-rate axis. The authors registered the band as $[-2.4, -1.6]$, to hold on at least three of four base models, and swept quantiser width $b \in \{1, 2, 3, 4, 8, 16\}$ with the fit over $b \in \{2, 3, 4, 8\}$, $b = 1$ being clipping-dominated. The rule voids the earlier falsification if the slope lies in the band on three of four bases, and lets it stand if $|\text{slope}| < 0.5$ on three of four.

First fault: a mismatched reference. An earlier version reported the prediction falsified on measured slopes of -0.10 and -0.21. **Those two numbers are withdrawn.** Each finite-rate curve was compared against a $b = \infty$ reference computed at a *different* ridge value and aggregated over a different number of cohorts, so the difference was not taken within one method. The symptom was visible in the output at the time and the authors did not act on it: on two bases the excess at finite rate came out *below* its own asymptote, which the model does not permit.

Second fault: neither surviving fit is supported at the precision the metric provides, and the published three-decimal table concealed it. The fits are on the residual above the asymptote, and those residuals are mostly tiny. Llama's three fitted points have residuals 0.0144, 0.0023 and 0.0000029 nats. The last is the $b = 8$ point and is what makes the fit possible at all: without it Llama has two usable points and is unfittable like Mistral and Qwen.

It sits about 350 times below this metric's stated resolution of 0.001 nats, so on a $\log_2$ axis it carries enormous leverage while being, by the authors' own account of the metric, indistinguishable from zero. Yi is worse in one respect: two of its three fitted residuals, 0.00046 and 0.000044, are below the resolution.

The authors also withdraw this section's earlier account of why Mistral and Qwen are unfittable. It said each "reaches its asymptotic value inside the fit window", which is benign and wrong. What happens is that their $b = 4$ and $b = 8$ points fall *below* their asymptotes and are removed by the residual filter: the same anomaly described above, which the published caption asserted no longer occurred. It occurs on all four bases, ten times in twenty-four measurements, and Table 16 now reports it.

The rule, applied as written. The registered rule needs the slope in $[-2.4, -1.6]$ on three of four bases to void the falsification, and $|\text{slope}| < 0.5$ on three of four for it to stand. The authors obtain one and zero, so the outcome is UNRESOLVED. The authors apply the rule to the fits as computed rather than adjusting the residual filter now that they can see which points it admits: moving a threshold after seeing its consequences is what the pre-registration exists to prevent, and here it would move the verdict in their own favour.

The outcome is unresolved for a more basic reason than the rule contemplates, though. The rule presumes fits worth judging, and on this grid there are none: only $b \in \{1, 2\}$ is comfortably above the noise floor on any base, and $b = 1$ is excluded as clipping-dominated. The registered window $\{2, 3, 4, 8\}$ is too coarse for bases that reach their noise floor by three bits. A finer low-rate grid would settle it, and the authors have deliberately not chosen one after seeing which grid would produce a fit.

What survives is operational. With the ridge on, three bits buys essentially the whole effect: from $b = 3$ upward every base is within 0.005 nats of its own asymptote, and sixteen bits adds nothing measurable over four. Only $b = 1$ is far off. A practitioner quantising a merged adapter can stop at three or four bits, and that recommendation does not depend on any exponent.

Caveats. This sweep is a single initialisation draw on the shared-initialisation cohort, so it says nothing about whether finite-rate behaviour differs by regime; repeating it on an independent cohort asks a different question from the one registered here and would need its own rule. Both faults were the authors' and both were in the analysis rather than the cells: the first in the reference the analyser selected, the second in the precision at which they reported and read the result. The second was invisible at three decimals and would have stayed invisible, which is why the table above prints the fitted points and the residual scale rather than rounded values alone.

F Pre-Registration Documents and Commit Ordering

This appendix records the pre-registration documents governing the tests in §5, together with the commit ordering that makes them checkable. In every case the decision rules were committed before the compute they govern was dispatched, and in the DARE-TIES case before the method being tested had been implemented at all.

Why the ordering is the evidence. A pre-registration that can be edited after the fact is worth nothing. The claim the authors are making is not that they intended to be honest, but that the repository history makes dishonesty detectable: each rules commit is an ancestor of every result commit it governs, and the rules files are not modified after their initial commit. A reader who does not trust the narrative can verify the ordering directly.

Table 17 gives the hashes, so that checking the ordering does not require reconstructing which commit is which. Each row reads left to right in the order the commits were made, and every rules commit is an ancestor of the result commit on its row.

Two hash spaces, and only one of them resolves. The pre-registration documents reproduced in §F.4 cite commits too, recording by hash that a document was committed before the generator it governs. Those are the hashes the authors' working repository had when each document was written, and they will *not* resolve in the published

Table 17. Commit trail for every pre-registered test in this paper. The hashes are those in the public repository, which differ from the authors' working repository's because the published history is filtered; the ordering is unaffected, since filtering does not change topology. Verify a row with `git merge-base --is-ancestor <rules> <result>`, which exits zero exactly when the ordering claimed here holds, and fails on the reverse.

test	rules	analyzer or implementation	result
Replication, step 0 (§5.1)	`947f0ea`	—	`1dcadd4`
Replication, $n = 3$ (§5.1)	`9b3f57e`	—	`4328446`
DARE with TIES (§5.10)	`5b58059`	`b9b1678`, `10bd1e4`	`f33cb02`
Conditioning and the ridge (§5.2)	`dc0a1f1`	`ac4499f`, `25d4d47`	`efa1d2d`
Rate exponent (§5.9)	`dc0a1f1`	`99fd95f` (corrected)	`c0a1cb3`
Solver replication (§5.4)	`876fcb6`	`2adbb79`, `1733865`	`57a8f89`
Untruncated and gate (§5.6)	`876fcb6`	`e4043b4`	`63bdf0c`
Repaired KnOTS (§6)	`876fcb6`	`45493e4`	`9f32ed0`
Merge matrix at $T = 3$ (§5.1)	`876fcb6`	`4c95967`	`908a486`
Margin-aware control (§5.1)	`876fcb6`	`d070a1a`	`0a4afaf`
Shared arm at $n = 3$ (§5.3)	`3b9db2f`	`7c25aa1`	`0a4afaf`
Public-cohort prevalence (§4.6)	`335a4f6`	`0b6eb40`	`0a4afaf`
Downstream accuracy (§5.8)	`a56b31e`	`321a808`	`0a4afaf`
Heuristics null in accuracy (§5.2)	`91d4a4b`	`4b5a56c`	`9b77f19`
Drift across training (App. D.4)	`8544b36`	`0c8770b`	`5e01813`

repository: filtering rewrites every commit, so the hash a document was written with no longer names anything. The authors have left them as written rather than editing the documents, because editing a pre-registration after the fact is the one thing these documents exist to make detectable, and a reader can see they are unedited precisely because their internal references are stale.

Only the hashes in Tables 17 and 18 resolve in the published repository, and those are the ones the verification asks you to use. The correspondence between the two spaces is recoverable: commit subjects survive filtering, so a stale reference can be matched by its message.

Three rows deserve a comment, and the first is the one a sceptical reader should check first.

The replication appears as two rows because its amendment does *not* precede the step-0 result: `9b3f57e` was committed after `1dcadd4` was read. That is the blindness limitation the amendment states about itself, made checkable rather than asserted, and it is why the amendment is paired with the $n = 3$ verdict it actually governs. Running the ancestry check on `9b3f57e` against `1dcadd4` fails, and should.

The DARE-TIES analyzer appears twice because the first version printed, under a mixed verdict, the consequence text belonging to a different verdict; `10bd1e4` is that repair, and it predates the verdict commit. The rate-exponent row names a *corrected* analyzer, and that correction was made after the first analysis had been read, for the reason given in §F.3 below.

F.1 Replication of the merge matrix on independent cohorts

Rules committed before any independent-cohort result was read. An amendment fixing how the three cohorts are combined was committed before `indep2` and `indep3` were read, and states its own blindness limitation in its first paragraph: `indep1` had already been read when the aggregation rule was chosen, so one of three inputs was visible. The authors flag this rather than claim full blindness.

The operative rules, summarised here and reproduced in full in the public repository:

- Unit of analysis is the cohort, $n = 3$.
- Tie threshold 0.005 nats, fixed in the parent document and not changed.
- Noise gate is one-directional: with $\mathrm{SE} = \mathrm{sd}/\sqrt{3}$, a win or loss is downgraded to a tie unless $|\mathrm{mean}(d)| > 2 \times \mathrm{SE}$. The gate may only downgrade, never promote.
- Primary comparison uses the champion baseline selected on the mean across cohorts, with two pre-specified robustness lines: champion re-selected within each cohort, and the original single-cohort rule applied three times with a majority vote.
- A control question asks how many base models have a non-unanimous top-1 method across the three cohorts. Two or more withdraws the initialisation-confounding claim.

The control returned two of four, and the claim was withdrawn (§5.7).

F.2 DARE composed with TIES

Rules committed before the `dare_ties` method existed in the codebase, which is the strongest ordering guarantee available: the implementation commit is a descendant of the rules commit, so the rules cannot have been written to fit an implementation that had not been written.

The operative rules, summarised here and reproduced in full in the public repository:

- 36 cells: four base models × three cohorts × DARE density in $\{0.5, 0.2, 0.1\}$, with the TIES trim density pinned at 0.2 in both arms so the only difference from the plain TIES arm is the DARE mask.
- The primary verdict is read at DARE density 0.2 only, named in advance. If another density flatters DARE, that is a secondary observation and is not promoted.
- Per base: helps if mean $d > 0.005$ and the gate passes, hurts if mean $d < -0.005$ and the gate passes, neutral otherwise. Overall: helps if it helps on ≥ 2 and hurts on none; hurts if the mirror image; neutral if neutral on ≥ 3; *mixed* otherwise, and then neither direction is claimed.
- The TIES reference cells are the pre-existing ones and are not re-run, re-seeded, or re-selected.
- A smoke test on one real cell is mandatory before the remaining thirty-five are dispatched.
- All three densities are reported, including any that flatter DARE and any that do not.

The outcome was mixed: helps on one base, hurts on three (§5.10).

Two disclosures. First, the mandated smoke cell sits at the primary density and is therefore one of the twelve cells deciding the primary verdict, so blindness for that test is eleven of twelve rather than twelve of twelve. The smoke inspection checked configuration equality between the arms and that the mask was not a no-op; it did not examine the direction of the effect. Second, the authors' analysis script initially printed, under a mixed verdict, the consequence text belonging to the hurts and neutral verdicts, which asserted that the negative result generalises. The pre-registration says of a mixed verdict only that neither direction is claimed. The verdict computation was correct against the rule; the reporting text was not, and it was corrected before any of it reached this paper.

F.3 Rate exponent

The band $[-2.4, -1.6]$ for the log-rate slope, the fit window $b \in \{2, 3, 4, 8\}$, and the requirement that the band hold on at least three of four base models were fixed before the sweep ran.

A second document was committed later, when an external review argued that the flat curve the authors were then reporting could be an artifact of the post-merge rank truncation. It fixed three branches in advance: the falsification is *void* if the slope enters the band on at least three of four bases, it *stands* if $|\mathrm{slope}| < 0.5$ on at least three of four, and it is *unresolved* otherwise, "reported as unresolved, not resolved in whichever direction is more convenient". That document also recorded, before any new cell ran, what to do about an anomaly already visible in the output: on two bases the excess at finite rate sat below the $b = \infty$ value, which the model forbids, and the rules said in advance that if it persisted it was to be treated as a harness problem and reported as one.

The experiment those rules were written for was never needed. Its premise was that the sweep had been truncated, and checking the configurations showed it had not been: the sweep already ran at the untruncated realisation, so there was nothing to repeat. The anomaly clause was the part that mattered, and following it led to the mismatched asymptote and the refit of §5.9. The authors applied the three branches to the refitted slopes as written, obtaining one base in band and none flat, hence UNRESOLVED.

Two things about this ordering deserve stating plainly. The decision rule is unchanged and predates the numbers it judges. But the rule was registered against a different intervention from the one that ended up producing those numbers, so what carried over is the standard of evidence rather than a full pre-specification of the analysis, and the correction itself was made after the first analysis had been read. The authors think a forbidden value obliges investigation rather than permitting it, which is why the clause was written in advance, and the commit ordering shows both the clause and what followed it.

F.4 The pre-registration documents

Twelve documents govern the tests in this paper. Table 18 lists them with the commit that introduced each, and the rules are summarised below. **The documents themselves are in the public repository, complete and unedited**, at the commits named here, which is the copy that cannot have been written afterwards. Earlier versions of this paper reproduced them all verbatim in the appendix; that is what the repository is for, and printing them twice made the paper longer without making anything more checkable.

Each is the file as committed and none was modified after its commit, which is checkable with `git log --follow` on the path. One character is altered across the twelve: the first document records the wall-clock time at which a set of cells completed, with a timezone abbreviation that would narrow the authors' geography under double-blind review, and it appears as `[tz]`. That is the only alteration.

Two conventions are shared by every document and are not repeated in the summaries: a **tie threshold of** 0.005 **nats**, fixed in the 2026-08-03 parent, and a **one-directional noise gate** at $2 \times$ SE over cohorts, which may downgrade a win or a loss to a tie but may never promote a tie to either. Verdicts are reported in whichever branch the rule lands in.

The rules, one paragraph each.

- **`prereg_a1_matrix`** (`947f0ea`) fixes the replication: does the encoder's win over the champion baseline survive the move to independently initialised cohorts, and do method rankings change between cohorts? It states its own statistical ceiling up front, that one cohort per arm cannot support a strong verdict regardless of what the rules return. Outcome: the win does not survive (§5.1).
- **`prereg_a1_matrix_amendment`** (`9b3f57e`) extends it to $n = 3$ cohorts and fixes the aggregation. It declares its own blindness status plainly: `indep1` had been read when the rule was written, so the rule was chosen to be the obvious one rather than selected from a menu. It also retires the parent's 0.013 nat provisional band.
- **`prereg_dare_ties`** (`5b58059`) asks whether the authors' DARE-with-task-arithmetic result extends to DARE with TIES, primary at density 0.2 only, with the density sweep explicitly secondary so that no density can be chosen after the fact. Outcome: it does not extend (§5.10).
- **`prereg_conditioning`** (`dc0a1f1`) governs the ridge sweep and the rate exponent: whether the optimal ridge tracks the conditioning, and whether the flat rate curve is a truncation artifact. It fixes the exponent band at $[-2.4, -1.6]$ on at least three of four bases. Outcomes: the ridge *gain* tracks conditioning while $\lambda^\star$ does not (§5.2), and the exponent is unresolved (Appendix E).

- **`prereg_tmlr`** (`876fcb6`) is the revision campaign: the go/no-go is whether cohort conditioning affects a solver the authors did not build, with the two cohorts required to match byte-for-byte on tasks, evaluation seed, sequence length and VRAM gate. It also governs the KnOTS repair and the $T = 3$ arm. Outcome: both predictions hold on four of four bases (§5.4).
- **`prereg_tmlr_amendment`** (`2adbb79`) sets RegMean's minimally regularised reference to $\lambda = 10^{-6}$ rather than 0, because RegMean solves in the full input dimension against a Gram of rank at most Tr, so $\lambda = 0$ is singular rather than ill-conditioned. It changes a grid point, not a threshold or a rule.
- **`prereg_shared_arm`** (`3b9db2f`) answers a limitation the draft raised against itself: the ridge sweep's shared arm was $n = 1$, so its gate was one-sided. This registers the shared arm at $n = 3$ with a genuinely two-sided gate, and flags in advance that the published cohort sitting more than 2 sd from its arm's mean would require reporting. Outcome: both predictions 4/4, and the published cohort sits 0.23 to 0.75 sd out (§5.3).
- **`prereg_prevalence`** (`335a4f6`) fixes the sampling frame for the public-cohort audit before any download, in three parts: the library condition (A), published benchmarks' released cohorts (B), and the broad sample (C). Outcome: A and C reported (§4.6), B not run.
- **`prereg_downstream`** (`a56b31e`) asks whether the conditioning collapse is visible in accuracy rather than only in NLL, on GSM8K and HumanEval, with a binomial-SE gate. It fixes in advance what the authors would write in each branch, including that a null would move the metric limitation into §1 and delete the practice recommendations. Outcome: the ridge recovers the shared arm on four of four bases on both benchmarks (§5.8).
- **`prereg_downstream_amendment`** (`5216230`) corrects the authors' own specification error: it registered $n = 500$ evaluation items, but HumanEval has 164. The gate adapts to the actual n, which widens it to roughly 11 points on HumanEval against 6 on GSM8K, so the two benchmarks are not equally powered and a null on HumanEval is not read as equivalent to a null on GSM8K. It makes the authors' own predictions harder to satisfy, and was written before any cell ran.
- **`prereg_drift`** (`8544b36`) discharges E3, which asked how far A travels from its initialisation across training and which the authors had reported as unevaluable because the original cohorts kept no initial weights and no intermediate checkpoints. It trains a fresh cohort with both, on one base model in both initialisation arms, and fixes three predictions: that A ends within half its own norm of where it started, that the trace is monotone, and that the cohort reproduces the collapse at all. Branch 3 is the one that would have cost the authors: if A travelled far and the subspaces still collapsed, the mechanism of §4.1 would be falsified and the abstract would have to say so. Outcome: all three hold (Appendix D.4).
- **`prereg_heuristics_downstream`** (`91d4a4b`) answers an asymmetry the authors found in their own reporting: the null of §5.2 is measured only in worst-task NLL excess, the unit §6 disclaims, while the positive result beside it has two units. It registers the same five heuristics on the same two cohorts, scored on GSM8K and HumanEval, 80 cells. Two features are worth naming because both cut against the authors. The gate is *binomial* rather than the $2 \times$ SE-over-cohorts gate used elsewhere, since this design has one cohort per arm, which makes it a weaker instrument with a minimum detectable effect of about 9 accuracy points on GSM8K and 16 on HumanEval; the document fixes the wording any null must be reported in so that it cannot later be read as "no effect". And it re-runs *both* arms rather than reusing the 40 shared-arm cells that already exist, because those were scored before the scorer repairs of §5.8 and a paired difference taken across two scorer versions would carry a known, method-dependent defect inside it. All four branches of the decision rule are written out in advance, and two of them remove or narrow contribution 3.

Table 18. The twelve pre-registration documents and the commit that introduced each. Every one was added in a single commit and never modified afterwards, which is checkable with `git log --follow` on the path.

document	introduced in	governs
`prereg_a1_matrix`	`947f0ea`	replication, step 0
`prereg_a1_matrix_amendment`	`9b3f57e`	replication, $n = 3$
`prereg_dare_ties`	`5b58059`	DARE with TIES
`prereg_conditioning`	`dc0a1f1`	ridge sweep, rate exponent
`prereg_tmlr`	`876fcb6`	solver replication, gate, KnOTS, $T = 3$
`prereg_tmlr_amendment`	`2adbb79`	RegMean's minimal λ
`prereg_shared_arm`	`3b9db2f`	the shared arm at $n = 3$
`prereg_prevalence`	`335a4f6`	public-cohort prevalence
`prereg_downstream`	`a56b31e`	downstream accuracy
`prereg_downstream_amendment`	`5216230`	HumanEval's 164 items
`prereg_heuristics_downstream`	`91d4a4b`	the heuristics null in accuracy
`prereg_drift`	`8544b36`	how far A travels (E3)

G Experiment Manifest

This appendix lists the experiments this paper reports, so that the released code and result files can be matched to the claims they support. Results built on the shared-initialisation cohort alone are not carried over as method conclusions, because §4 shows that cohort sits in a different regime and §5.1 shows conclusions drawn there do not transfer. Where that cohort appears below it is as one arm of a paired comparison, never on its own.

Cohorts and adapters. Four base models (Llama-3.1-8B, Mistral-7B-v0.3, Qwen2.5-7B, Yi-1.5-9B-Chat), four tasks per cohort at LoRA rank $r = 16$, so $T = 4$ and $Tr = 64$. Four cohorts per base: one *shared* cohort in which all four adapters are initialised from a single global seed, and three *independent* cohorts (`indep1`, `indep2`, `indep3`) in which each task draws its own initialisation. That is $4 \times 4 \times 4 = 64$ trained adapters. Training data and data-ordering seed are identical across cohorts, so the only variable is the initialisation draw.

Table 19. Experiments reported in this paper. Cell counts are completed evaluation cells, excluding adapter training.

experiment	where	driver	cells
Subspace geometry audit	§4.2, App. D	`measure_subspace_geometry.py`	16 (CPU)
Merge matrix, independent cohorts	§5.1	orchestrator	84
Merge matrix, shared cohort (the null)	§5.2	orchestrator	20
Ridge sweep, both cohorts	§5.2	`run_eval_cell.py`	56
Rate sweep, quantiser width b	§5.9	`run_eval_cell.py`	24 + 4
DARE density sweep	§5.10	`run_eval_cell.py`	20
DARE-TIES composition	§5.10	orchestrator	36
Synthetic validation	App. C	`run_synthetic.py`	CPU

Metric. Worst-task negative log-likelihood excess in nats, relative to the per-task fine-tuned adapter, evaluated on a held-out draw with the evaluation seed pinned across every arm of every comparison. The merged model is truncated back to rank r after merging for every method, so no method gains capacity over another.

Methods compared. Task arithmetic (Ilharco et al. 2023), TIES (Yadav et al. 2023), DARE (Yu et al. 2024), DARE composed with TIES, task-vector quantisation (Kim et al. 2025), KnOTS (Stoica et al. 2025), data-free RegMean (Jin et al. 2023), and the authors' rate-distortion encoder with and without the Tikhonov ridge.

KnOTS appears in two forms and the distinction matters. Under its default inner combination the authors' implementation reduces algebraically to task arithmetic ($VV^\top = I$ cancels), so it was not measuring what they intended; that version is excluded from any claim about subspace alignment (§6). The repaired version, with a TIES inner combination, is the one carried in the regime null of §5.2, in the 80-cell accuracy replication of the same null, and in the five methods §1 names. Where this paper says "a repaired KnOTS" it means the second. The authors make no claim about the reference implementation of published KnOTS, which they did not run.

RegMean (Jin et al. 2023) is run here in its data-free adapter-only form on both cohorts, on the same grid and under a registered rule (§5.4), and its published α-shrinkage default is measured separately (§5.5). That is what licenses talking about the closed-form least-squares family rather than about the authors' encoder alone, and it is the second solver Contribution 3 rests on.

Data-free AdaMerging (Yang et al. 2024) is *not* carried over. It was run on the shared-initialisation cohort only, and §5.1 is the authors' reason for not transferring conclusions from that cohort. LoRM and RegMean++ are not run either, for the reasons in §2: neither has a data-free adapter-only form the authors could implement without inventing the part that makes it that method. So the family claim rests on **two** members, the authors' encoder and RegMean, and not on five.

Geometry audit cost. The audit of §4.5 reads the adapter weight files, computes principal angles between the T task row spaces over eight sampled layers, and returns soft and hard effective dimension with the implied floor. It runs on CPU in about one minute per cohort, needs no evaluation data and performs no forward passes, which is the property that makes it usable as a pre-merge check.

Compute. All GPU work ran on a single node with A100-80GB cards, one evaluation cell per card, each holding 18 to 21 GiB. Cells are dispatched by an orchestrator that records a per-cell done marker, so an interrupted run resumes without recomputing completed cells.

H Reproducibility Checklist for JAIR

Select the answers that apply to your research – one per item.

All articles:

(1) All claims investigated in this work are clearly stated. **[yes]**
(2) Clear explanations are given how the work reported substantiates the claims. **[yes]**
(3) Limitations or technical assumptions are stated clearly and explicitly. **[yes]**
(4) Conceptual outlines and/or pseudo-code descriptions of the AI methods introduced in this work are provided, and important implementation details are discussed. **[yes]**
(5) Motivation is provided for all design choices, including algorithms, implementation choices, parameters, data sets and experimental protocols beyond metrics. **[yes]**

Articles containing theoretical contributions:

Does this paper make theoretical contributions? **[yes]**

If yes, please complete the list below.

(1) All assumptions and restrictions are stated clearly and formally. **[yes]**
(2) All novel claims are stated formally (e.g., in theorem statements). **[yes]**

(3) Proofs of all non-trivial claims are provided in sufficient detail to permit verification by readers with a reasonable degree of expertise (e.g., that expected from a PhD candidate in the same area of AI). **[yes]**
(4) Complex formalism, such as definitions or proofs, is motivated and explained clearly. **[yes]**
(5) The use of mathematical notation and formalism serves the purpose of enhancing clarity and precision; gratuitous use of mathematical formalism (i.e., use that does not enhance clarity or precision) is avoided. **[yes]**
(6) Appropriate citations are given for all non-trivial theoretical tools and techniques. **[yes]**

Articles reporting on computational experiments:

Does this paper include computational experiments? **[yes]**

If yes, please complete the list below.

(1) All source code required for conducting experiments is included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for source code readability and documentation as well as for long-term accessibility. **[yes]**
(2) The source code comes with a license that allows free usage for reproducibility purposes. **[yes]**
(3) The source code comes with a license that allows free usage for research purposes in general. **[yes]**
(4) Raw, unaggregated data from all experiments is included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for long-term accessibility. **[yes]**
(5) The unaggregated data comes with a license that allows free usage for reproducibility purposes. **[yes]**
(6) The unaggregated data comes with a license that allows free usage for research purposes in general. **[yes]**
(7) If an algorithm depends on randomness, then the method used for generating random numbers and for setting seeds is described in a way sufficient to allow replication of results. **[yes]**
(8) The execution environment for experiments, the computing infrastructure (hardware and software) used for running them, is described, including GPU/CPU makes and models; amount of memory (cache and RAM); make and version of operating system; names and versions of relevant software libraries and frameworks. **[partially]**
(9) The evaluation metrics used in experiments are clearly explained and their choice is explicitly motivated. **[yes]**
(10) The number of algorithm runs used to compute each result is reported. **[yes]**
(11) Reported results have not been "cherry-picked" by silently ignoring unsuccessful or unsatisfactory experiments. **[yes]**
(12) Analysis of results goes beyond single-dimensional summaries of performance (e.g., average, median) to include measures of variation, confidence, or other distributional information. **[yes]**
(13) All (hyper-) parameter settings for the algorithms/methods used in experiments have been reported, along with the rationale or method for determining them. **[yes]**
(14) The number and range of (hyper-) parameter settings explored prior to conducting final experiments have been indicated, along with the effort spent on (hyper-) parameter optimisation. **[partially]**
(15) Appropriately chosen statistical hypothesis tests are used to establish statistical significance in the presence of noise effects. **[yes]**

Articles using data sets:

Does this work rely on one or more data sets (possibly obtained from a benchmark generator or similar software artifact)? **[yes]**

If yes, please complete the list below.

(1) All newly introduced data sets are included in an online appendix or will be made publicly available upon publication of the paper. The online appendix follows best practices for long-term accessibility with a license that allows free usage for research purposes. **[NA]**
(2) The newly introduced data set comes with a license that allows free usage for reproducibility purposes. **[NA]**
(3) The newly introduced data set comes with a license that allows free usage for research purposes in general. **[NA]**
(4) All data sets drawn from the literature or other public sources (potentially including authors' own previously published work) are accompanied by appropriate citations. **[yes]**
(5) All data sets drawn from the existing literature (potentially including authors' own previously published work) are publicly available. **[yes]**
(6) All new data sets and data sets that are not publicly available are described in detail, including relevant statistics, the data collection process and annotation process if relevant. **[NA]**
(7) All methods used for preprocessing, augmenting, batching or splitting data sets (e.g., in the context of hold-out or cross-validation) are described in detail. **[yes]**

Explanations on any of the answers above (optional):

Computational experiments, items 1 and 4: which clause the authors satisfy. They provide no online appendix. Both items are met by their second clause: the source code and the raw per-cell result files are already public, at https://github.com/K144U/Initialisation-Sets-the-Conditioning-of-LoRA-Merging, rather than deposited with this paper. That is deliberate. What Appendix F asks a reader to check is the *order* in which things were committed, and only a repository with its commit graph intact supports that: the reader runs `git merge-base` and gets an answer. A static archive would turn the same claim back into something taken on trust, which is the one thing this paper is trying not to ask for.

Data sets, items 1–3 and 6. This work introduces no new data set. All four training tasks and both downstream benchmarks are public and cited.

Computational experiments, item 8: partially. The GPU make and model and the per-cell memory footprint are reported in the Reproducibility Statement, as is the aggregate compute. Operating system version and pinned library versions are recorded in the released environment files rather than in the paper.

Computational experiments, item 14: partially. The regularisation sweep reports its full grid and range. The authors do not report a systematic accounting of settings explored before the final experiments, because most were not hyperparameter search: they were superseded runs discarded after defects were found in the scorers and in one data loader, and those are described in §6 rather than summarised as tuning effort.

Item 11, on cherry-picking. Every empirical claim in this paper is governed by a pre-registration committed to version control before the compute it governs was dispatched, with all outcome branches written in advance. The registrations are included unedited and the commit ordering is verifiable from the released history with `git merge-base`. Several of the authors' own earlier claims did not survive these tests and are reported as withdrawn rather than removed.